

Brainstem neurons coordinate the bladder and urethra sphincter for urination

Reviewed Preprint

v1 • December 3, 2024

Not revised

Xing Li, Xianping Li, Jun Li, Han Qin, Shanshan Liang, Jun Li, Tingliang Jian, Xia Wang, Lingxuan Yin, Chunhui Yuan, Xiang Liao, Hongbo Jia ✉, Xiaowei Chen ✉, Jiwei Yao ✉

Advanced Institute for Brain and Intelligence, School of Physical Science and Technology, Guangxi University, Nanning, China • Brain Research Center and State Key Laboratory of Trauma and Chemical Poisoning, Third Military Medical University, Chongqing, China • Department of Urology, PLA Naval Medical Center, Naval Medical University, Shanghai, China • LFC Laboratory and Chongqing Institute for Brain and Intelligence, Guangyang Bay Laboratory, Chongqing, China • Center for Neurointelligence, School of Medicine, Chongqing University, Chongqing, China • Leibniz Institute for Neurobiology, Magdeburg, Germany • Suzhou Institute of Biomedical Engineering and Technology, Chinese Academy of Sciences, Suzhou, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

In this **important** study, Li et al. identify estrogen receptor 1-expressing neurons (ESR1+) in Barrington's nucleus as key regulators coordinating both bladder contraction and the relaxation of the external urethral sphincter. Using appropriate and validated methodologies aligned with the current state of the art, the data are **convincing** and of generally high quality.

<https://doi.org/10.7554/eLife.103224.1.sa4>

Abstract

Urination, a vital and conserved process of emptying urine from the urinary bladder in mammals, requires precise coordination between the bladder and external urethra sphincter (EUS) that is tightly controlled by a complex neural network. However, the specific subpopulation of neurons that accounts for such coordination remains unidentified, limiting the development of target-specific therapies for certain urination disorders, e.g. detrusor-sphincter dyssynergia. Here, we find that cells expressing estrogen receptor 1 (ESR1⁺) in the pontine micturition center (PMC) initiate voiding when activated and suspend ongoing voiding when suppressed, each at 100% reliability, respectively. Transection of either the pelvic or the pudendal nerve does not impair PMC^{ESR1+} control of the downstream target through the other nerve at all. Anatomically, PMC^{ESR1+} cells possess two subpopulations projecting to either the pelvic or pudendal nerve and a third, dual-projecting subpopulation, locking in the coordination of bladder contraction and sphincter relaxation in a rigid temporal order. We identify a cell type in the brainstem that controls the bladder-urethra coordination for urination.

Introduction

Urination is a basic life-maintaining function involving the coordinated control of two functional units of the lower urinary tract (Andersson and K.-E., 2004; Chang et al., 2018; Fowler et al., 2008), namely the bladder detrusor and external urethral sphincter (EUS). The coordination between the bladder and urethra sphincter for initiating or suspending voiding whenever needed should be executed at 100% reliability in healthy subjects. By contrast, impairment of such coordination (Cho et al., 2015; Taweel et al., 2015; Griffiths et al., 2005), at any rate, leads to various lower urinary tract dysfunctions (Sakakibara and Ryuji, 2015; Drake et al., 2014), significantly degrading the quality of life (Milsom et al., 2014; Aoki et al., 2017). While individual neural control pathways to either the bladder detrusor or the urethral sphincter have been extensively studied (Xiao et al., 2021; Lee et al., 2021), little is known about the neural mechanisms underlying their coordination, which impedes progress in developing target-specific therapies for certain urination disorders, such as detrusor-sphincter dyssynergia (DSD) (Stoffel, 2016; Seseke et al., 2019).

Individual neural pathways are present at different levels of the brain for the control of either the bladder or the EUS (Malykhina, 2017; Jin et al., 2020; Mukhopadhyay and Stowers, 2020). For example, at the cortical level, a subset of motor cortex neurons has been found to drive bladder contraction through the projection to the brainstem (Yao et al., 2018). At the brainstem level, the pontine micturition center (PMC; also referred to as Barrington's nucleus) has long been considered a command hub region for urination control (Benarroch, 2010; Morrison, 2008). More recent studies have demonstrated that neurons expressing corticotropin-releasing hormone (CRH) in the PMC (PMC^{CRH+}) control bladder contraction (Hou et al., 2016; Ito et al., 2020), and neurons expressing estrogen receptor 1 (ESR1) in the PMC (PMC^{ESR1+}) relax the sphincter (Keller et al., 2018), respectively. In addition, other subcortical regions, including the periaqueductal gray (PAG) (Rao et al., 2022), lateral hypothalamic area (LHA) (Verstegen et al., 2019; Hyun et al., 2021), and medial preoptic area (MPOA) (Hou et al., 2016), have been suggested to play a potential role in urination control by sending direct projections to the PMC. However, which brain areas coordinate both the bladder and urethra sphincter remains unclear.

The most likely candidate region for coordinated control is the PMC. Neural tracing experiments demonstrate that the PMC directly sends two bundles of glutamatergic axonal projections, one to the sacral parasympathetic nucleus (SPN), where parasympathetic bladder motoneurons are located, which send axons through the pelvic nerves for bladder control; and the other one to the lumbosacral dorsal gray commissure (DGC) interneurons, which inhibit the sphincter motoneurons in the dorsolateral nucleus (DL) for sphincter control via the pudendal nerves (Kawatani et al., 2021; Jin et al., 2020). Early studies show that microinjection of drugs into the PMC or electrical stimulation in the PMC causes bladder contraction, sphincter relaxation, and urination (Mallory, 1991; Noto et al., 1989; Sugaya and De Groat, 1994). However, the PMC consists of molecularly disparate cell subtypes characterized by the expression of different marker genes, e.g. CRH positive cells (Ito et al., 2020; Van Batavia et al., 2021), ESR1 positive cells (Vanderhorst et al., 2005), vesicular glutamate transporter (Vglut2) and vesicular GABA transporter (VGAT) positive cells (Verstegen et al., 2017; Hou et al., 2016). What remains unknown to date is which exact neuronal subpopulation in the PMC accounts for such coordination of the bladder and urethra sphincter. As previous anatomical results have shown that PMC^{CRH+} neurons mainly project to the SPN in the spinal cord, whereas PMC^{ESR1+} neurons project to both the SPN and DGC in the spinal cord (Keller et al., 2018; Kawatani et al., 2021), we hypothesize that PMC^{ESR1+} cells could be a candidate for the coordination control and began our investigation.

Results

Voiding tightly correlates with PMC^{ESR1+} cell activity

We performed fiber photometry monitoring of neuronal population Ca²⁺ activity (Rao et al., 2022) (fluorescence indicator: GCaMP6f) in the PMC of awake, unrestrained ESR1-Cre mice (Figure 1A and B, see Methods for detail). Each detected voiding event tightly correlated with a detected Ca²⁺ transient (Ca²⁺ transient preceding voiding by 0.7/0.3-0.9 s, median/25%-75% percentile, same notation hereinafter; n = 260 voiding events from 9 mice; Figure 1C and D), as verified by both an analysis of temporally shuffled data (peak $\Delta f/f$ value, data: 31.8%/26.7%-33.2%; shuffled: 5.4%/3.7%-6.0%, n = 260 events of 9 mice, $p = 3.9 \times 10^{-3}$, Wilcoxon signed-rank test; Figure 1E) and a blank control with an expression of inert fluorescence indicator, EYFP (Enhanced Yellow Fluorescent Protein), in PMC^{ESR1+} cells (detected voiding-associated fluorescence signal events: 100%/100% – 100%, n = 9 mice in test group; 0%/0% – 0%, n = 9 mice in control group, $p = 4.1 \times 10^{-5}$, Wilcoxon rank-sum test; Figure 1F-figure supplement 1). To further test single-neuron correlates of voiding, we performed an ‘opto-tagging’ experiment (Qin et al., 2022) with a tetrode-fiber bundle implanted in the PMC of ESR1-Cre mice injected with AAV2/8-DIO-ChR2-mCherry (Figure 1G; see Methods for detail). Single PMC^{ESR1+} units were sorted and tagged by detecting reliable spikes to brief optical stimulation (e.g., latency: 4.04/3.02-4.6 ms, success rate: 97.5%/91.5%-100%; Figure supplement 2). The baseline-corrected firing rate of PMC^{ESR1+} units significantly exceeded that of non-PMC^{ESR1+} units in the temporal association window with voiding (PMC^{ESR1+} units: ‘before’, 0.04/0.5-0.5 Hz, ‘voiding’, 1.4/2.5e-11-3.8 Hz, n = 28 cells, $p = 0.004$; non-PMC^{ESR1+} units: ‘before’, 0.8/0.2-1.5 Hz, ‘voiding’, 1/0.9-1.9 Hz, n = 51 cells, $p = 0.6$; Wilcoxon signed-rank test; Figure 1H and I). These results suggest that PMC^{ESR1+} cells’ firing activities are tightly associated with voiding.

Additionally, we performed a set of combined physiological monitoring experiments, integrating fiber photometry, cystometry, and electromyography (EMG) of external urethral sphincter (EUS) simultaneously in urethane-anesthetized mice (Figure supplement 3A, see methods for detail). For quantitative measurement, we applied cystometry by continuously infusing saline into the bladder through a microcatheter (30-50 μ l/min) to induce regular reflective voiding. A triple correlation of events was routinely observed, i.e., PMC^{ESR1+} neuronal activity, bladder pressure elevation, and the bursting pattern of EUS-EMG driving sphincter relaxation (Figure supplement 3B and C). This result was further validated by shuffled data analysis (Figure supplement 3D and E), demonstrating the robustness and time-locking precision of the triple correlation events. However, it is important to note that the association between PMC^{ESR1+} cell activity and voiding in the other way around was not always 100%, i.e., for each detected Ca²⁺ transient, there could be either a voiding contraction (VC) or occasionally a non-voiding contraction (NVC) (Bialosterski et al., 2011) (27.1% of all detected events were in such case; Figure supplement 4A-C). Nevertheless, not only was the peak of bladder pressure lower, but also the amplitude of the photometry Ca²⁺ transient, which is known as a reliable report of collective neuronal population activity level, was significantly lower in NVC events than in VC events (NVCs: 8.8%/6.6%-14.7%, n = 62 events from 3 mice; VC: 15.01%/10.3%-26.4%, n = 167 events from 3 mice, $p = 6.8 \times 10^{-9}$, Wilcoxon rank-sum test; figure supplement 4D and E), suggesting that not only the timing but also the strength of PMC^{ESR1+} cell activities were tightly correlated with successful voiding.

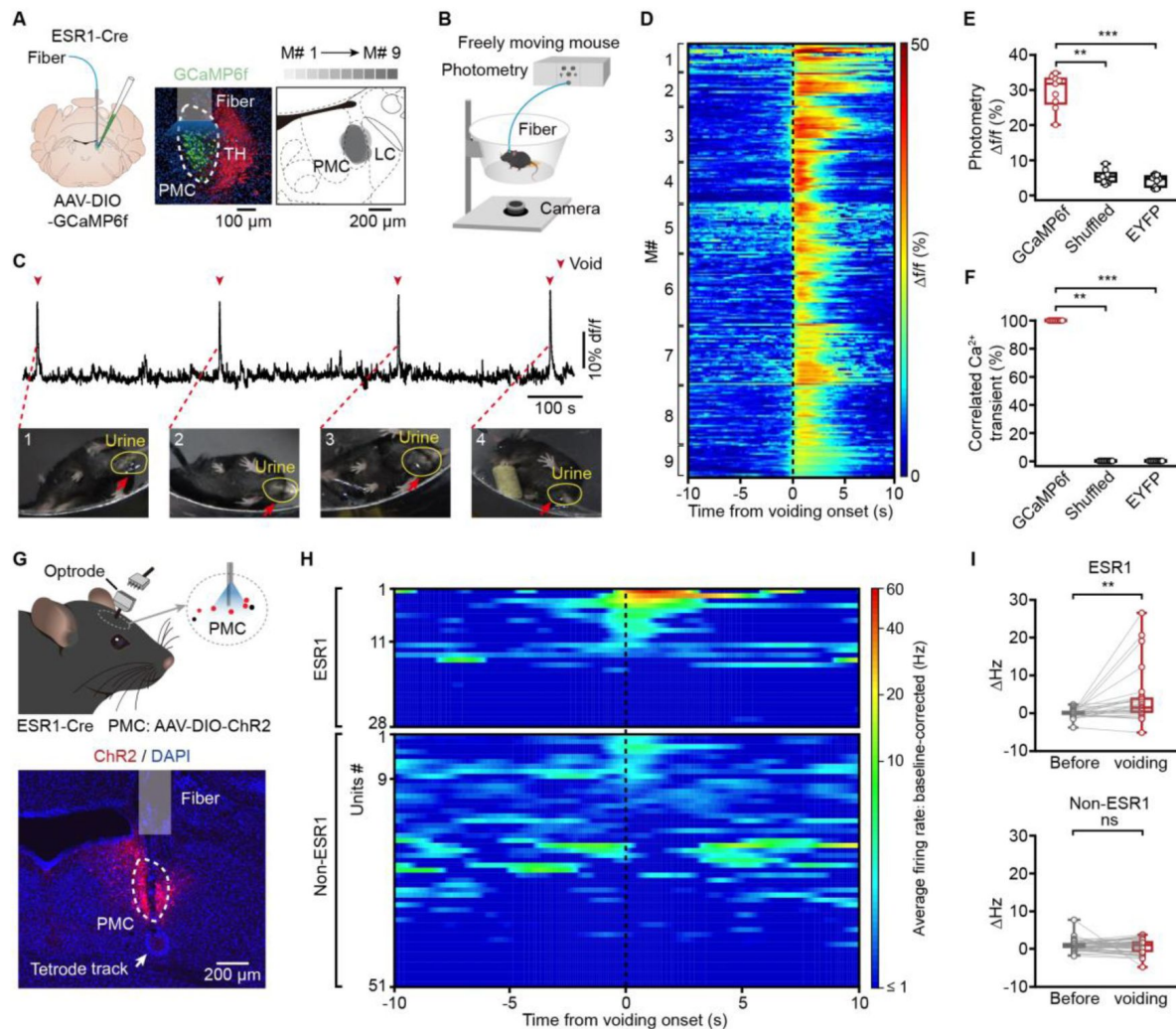


Figure 1.

PMCESR1⁺ cells tightly correlate with successful voiding.

(A) Schematic (left) of labeling and representative histology (middle) of PMCESR1⁺ cells labeled with GCaMP6f. Scale bar: 100 μ m. TH (tyrosine hydroxylase) stained neurons in the locus coeruleus. Right: Overlay of GCaMP6f-labeled areas from 9 mice. Scale bar: 200 μ m. (B) Schematic of fiber photometry Ca²⁺ recording in a freely moving mouse. (C) Representative Ca²⁺ traces (top) and voiding events (bottom, yellow circle). (D) Cumulative sessions of Ca²⁺ signals aligned to voiding onset (dotted line). (E) Boxplots showing the amplitude of voiding-related Ca²⁺ signals in various groups (n = 9 mice per group, 31.8% [26.7%-33.2%] for Gcamp6f, 5.4% [3.7%-6.0%] for Shuffled, 4.7% [2.5%-5.7%] for EYFP, **P = 3.9e-3 (Gcamp6f versus Shuffled), Wilcoxon signed-rank test; ***P = 4.1e-5 (Gcamp6f versus EYFP), Wilcoxon rank-sum test). (F) Detected voiding-related Ca²⁺ events in various groups (n = 9 mice per group, 100% [100%-100%] for Gcamp6f, 0% [0%-0%] for Shuffled, 0% [0%-0%] for EYFP. **P = 3.9e-3 (Gcamp6f versus Shuffled), Wilcoxon signed-rank test; ***P = 4.1e-5 (Gcamp6f versus EYFP), Wilcoxon rank-sum test). (G) Schematic (top) and representative histology (bottom) of optrode recording in the PMC of an ESR1-cre mouse. Scale bar: 200 μ m. (H) Cumulative sessions of sorted single-unit activity of PMCESR1⁺ (upper; n = 28 cells from 4 mice) and non-PMCESR1⁺ cells (lower; n = 51 cells from 4 mice) aligned to voiding onset (black dashed line), vertically arranged by their instantaneous firing rate at the voiding onset. (I) Boxplots showing the baseline-corrected average firing rates before and during voiding among PMCESR1⁺ (top, n = 28 cells from 4 mice, **P = 0.004) and non-PMCESR1⁺ cells (bottom, n = 51 cells from 4 mice, P = 0.6; n.s., not significant; Wilcoxon signed-rank test). For all data points in (E, F), and (I), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

PMC^{ESR1+} cells bidirectionally operate the bladder and sphincter to initiate or suspend voiding

With the tight correlation established, we moved on to test the causal relation between PMC^{ESR1+} cell activity and voiding. We started with a ‘loss-of-function’ test in awake mice, i.e., acute photoinhibition of PMC^{ESR1+} cells in a closed-loop to trigger the photoinhibition light as soon as the first patch of urine visualized (Figure 2A-figure supplement 5A; see Methods for detail). This test was also accompanied by a blank control in which all experimental conditions were the same except that the inhibitory opsin GtACR1 was absent. In photoinhibition events but not in the blank control events, the urine spot area was significantly reduced (‘Pre’: 34.1\27.1-40.4 cm²; ‘On’: 9.9\7.9-11.1 cm², n = 12 mice in test group, $p = 4.9\text{e-}4$; ‘Pre’: 35.2\27.9-43.9 cm²; ‘On’: 38.9\29.98-43.6 cm², n = 8 mice in control group, $p = 0.3$; Wilcoxon signed-rank test; **Figure 2B** and **C**), and the urination duration was significantly reduced as well (‘Pre’: 5.6\5.1-6.6 s; ‘On’: 1.38\1.19-1.42 s, $p = 4.9\text{e-}4$, Wilcoxon signed-rank test; **Figure 2D** and **E**). Effectively, the ongoing urination was fully suspended after a latency of 0.3 ± 0.1 sec from the onset of photoinhibition (**Figure supplement 5B**). To understand the physiological process during photoinhibition of PMC^{ESR1+} cells, we performed a simultaneous recording by measuring both the bladder pressure and the electromyograph of the external urethral sphincter (EUS-EMG) under urethane anesthesia (**Figure 2F**; see Methods for detail). This experiment also involved a closed-loop operation to trigger the light when observing the onset of the phasic bursting activity of the EUS-EMG that is known to be directly associated with successful voiding in rodents (Kadekawa et al., 2016a; Langdale and Grill, 2016). The maximum relative change in bladder pressure (Δ Pressure) upon voiding event was significantly reduced during photoinhibition compared to pre-inhibition baseline events (‘Pre’: 11.6\10.4-15.4 cm H₂O, ‘On’: 8.1\5.9-9.4 cm H₂O, n = 8 mice, $p = 7.8\text{e-}3$, Wilcoxon signed-rank test), but no such reduction was observed in the blank control group (‘Pre’: 10.7\6.9-12.03 cm H₂O, ‘On’: 10.5\7.2-13.4 cm H₂O, n = 7 mice, $p = 0.2$, Wilcoxon signed-rank test; **Figure 2G** and **H**). In the meanwhile, photoinhibition of the PMC^{ESR1+} cells also halted the voiding-associated, phasic bursting activity of EMG, effectively reducing the sphincter bursting duration (‘Pre’: 4.1\3.2-4.9 s, ‘On’: 0.98\0.9-1.2 s, $p = 7.8\text{e-}3$, Wilcoxon signed-rank test; **Figure 2G** and **I**) at a latency of 0.1 ± 0.02 sec (**Figure supplement 5C** and **D**), but no such effect was observed in the blank control group (‘Pre’: 3.3\2.9-4.8 s, ‘On’: 3.4\2.96-5.4 s, $p = 0.8$, Wilcoxon signed-rank test). Furthermore, the immediate suspension of an ongoing voiding event by photoinhibition of PMC^{ESR1+} cells resulted in an expected side effect that the threshold of bladder pressure to initiate the next voiding event (post-photoinhibition) became higher (**figure supplement 5E**). A control test for the above set of ‘loss-of-function’ experiments was to test whether shorter durations of photoinhibition also had the same urination suspension effect. To address this, we performed additional sets of control experiments in which a shorter duration of photoinhibition (5 s light-on, instead of 60 s) resulted in the same, 100% suspension effect (**Figure supplement 6**). These data together reveal that the acute photoinhibition of PMC^{ESR1+} cells halted both bladder contraction and sphincter relaxation, thereby leading to a full suspension of the ongoing voiding process.

Then, we performed a ‘gain-of-function’ test in awake mice, i.e., acute photoactivation of PMC^{ESR1+} cells that expressed the excitatory opsin channelrhodopsin-2 (ChR2), in a semi-closed loop to trigger the light on when the bladder was filled to a level that was lower than the threshold required for spontaneous voiding reflex (light on at 3.1 ± 0.03 min after the previous voiding; inter-voiding interval under control condition: 6.6 ± 0.5 min; **Figure 3A**; see Methods for detail). This test was also accompanied by a blank control in the absence of ChR2. Light stimulation initiated voiding event at 100% reliability (100%\100%-100%, n = 172 trials of 8 mice; **Figure 3B** and **C**) in the test group (response latency, 0.7\0.6-0.8 s; **Figure 3D**), but almost 0% (0%\0%-1.25%, n = 166 trials of 8 mice) in the blank control group (**Figure 3B** and **C**). Accordingly, we also performed simultaneous cystometry with the EUS-EMG experiment under urethane anesthesia (**Figure 3E**; see Methods for detail). Light stimulation induced a prominent upstroke of bladder pressure in the test group and did not affect bladder pressure at all in the

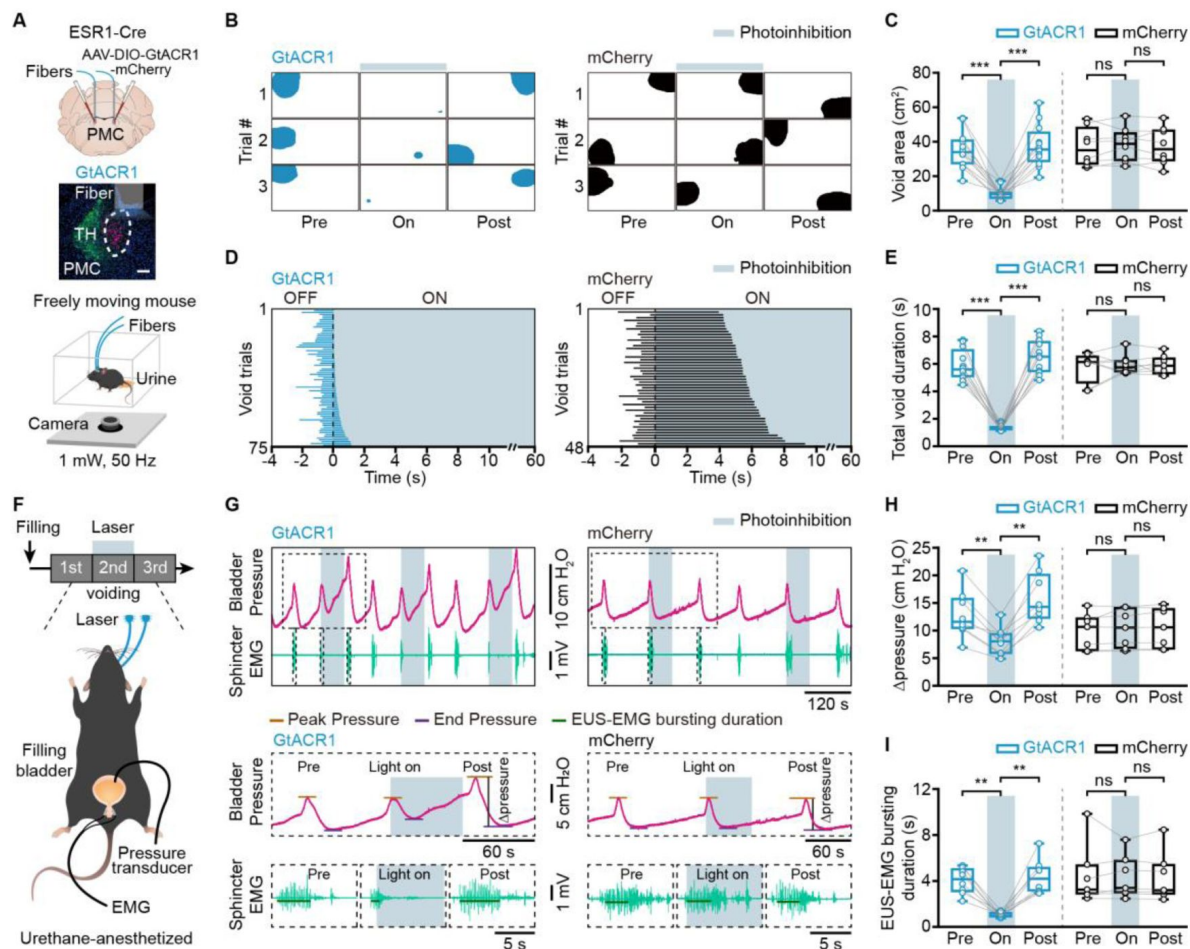


Figure 2.

Inactivation of PMCESR1⁺ cells suppresses bladder contraction and sphincter relaxation to suspend voiding.

(A) Schematic of labeling (top), representative histology (middle), and behavior test (bottom) for PMCESR1⁺ photoinhibition. Scale bar: 100 μ m. (B, C) Representative images (B) and quantification (C) of the void area before ('Pre'), during ('On'), and after ('Post') photoinhibition in PMCESR1-GtACR1 (n = 12 mice) and PMCESR1-mCherry (n = 8 mice) groups (from left to right: *** P = 4.9e-4, *** P = 4.9e-4, P = 0.3, P = 0.5, respectively; n.s., not significant; Wilcoxon signed-rank test). (D) Cumulative trials of voiding duration in PMCESR1-GtACR1 (blue bar, n = 75 trials from 12 mice) and PMCESR1-mCherry (black bar, n = 48 trials from 8 mice) during photoinhibition. Voiding trials are ordered by the increasing time of the voiding epoch with the laser on. (E) Voiding duration before, during, and after photoinhibition in PMCESR1-GtACR1 and PMCESR1-mCherry groups (from left to right: *** P = 4.9e-4, *** P = 4.9e-4, P = 0.7, P = 0.9, respectively; n.s., not significant; Wilcoxon signed-rank test). (F) Timeline (top) and schematic (bottom) for PMCESR1⁺ photoinhibition during simultaneous cystometry and electromyography recording. (G) Representative traces (top) and expanded portions (bottom) of bladder pressure (magenta) and EUS-EMG (teal) before, during, and after photoinhibition in PMCESR1-GtACR1 (left) and PMCESR1-mCherry (right) groups. (H, I) Quantification of the Δ pressure (H) and EUS-EMG bursting duration (I) during voiding before, during, and after photoinhibition in PMCESR1-GtACR1 (n = 8 mice) and PMCESR1-mCherry (n = 7 mice) groups (H: from left to right: ** P = 7.8e-3, ** P = 7.8e-3, P = 0.2, P = 0.3, respectively; I: from left to right: ** P = 7.8e-3, ** P = 7.8e-3, P = 0.8, P = 0.4, respectively; n.s., not significant; Wilcoxon signed-rank test). For all data points in (C, E, H), and (I), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

control group (ΔP : 5.6\4.5-8.2 cm H₂O, n = 9 mice in the test group, -0.1\0.2-0.1 cm H₂O, n = 6 mice in the control group, $p = 4e-4$, Wilcoxon rank-sum test; **Figure 3F-H**). Meanwhile, light stimulation triggered the phasic bursting activity of EUS-EMG, at a success rate of 100% in the test group and nearly 0% in the control group (**Figure 3F-H**). An additional set of control experiments by using regular inter-stimulation intervals (5 s light-on per every 30 s) instead of the threshold-adaptive interval, yielded the same 100% success rate of both bladder pressure and EUS-EMG responses (**Figure supplement 7**). Thus, photoactivation of PMC^{ESR1+} cells resulted in initiating the voiding process through both contracting the bladder and relaxing the sphincter. These sets of ‘loss-of-function’ and ‘gain-of-function’ experiments together demonstrate that PMC^{ESR1+} cells perform as a 100% reliable ‘master switch’ for either initiate or suspend voiding, provided that all downstream targets are directly controlled and operational, which will be tested next.

Transection of either the pudendal or pelvic nerve does not impair PMC^{ESR1+} neuronal control of the other

To further confirm whether the effect of PMC^{ESR1+} cells on the bladder can occur independently of sphincter relaxation, we designed a new set of simultaneous cystometry and EUS-EMG experiments with PMC^{ESR1-ChR2} mice subjected to first a pudendal nerve transection (PDNx, disrupting the pathway from the PMC to the EUS) and then additionally a pelvic nerve transection (PLNx, disrupting the pathway from the PMC to the bladder) in the same mice under urethane anesthesia (**Figure 4A**; see Methods for detail). Under PDNx condition, PMC^{ESR1+} cell photoactivation could not elicit EUS-EMG response at all, but robustly elicited bladder pressure upstroke which was then fully abolished after PLNx (**Figure 4B-E**). Notably, the photoactivation-induced bladder pressure upstroke under the PDNx condition was nearly the same as that in the pre-transection control condition (**Figure 4D**), suggesting that the PMC^{ESR1+} cell control of the bladder was fully operational even when the pudendal nerve to sphincter was severed.

Given that reflex signals from bladder afferents can indirectly influence the urethral sphincter activity (Chang et al., 2007), we next investigated whether activation of PMC^{ESR1+} cells directly induces the bursting pattern of EUS in the absence of bladder contraction. To this end, the sequential nerve transection test was performed again with another group of PMC^{ESR1-ChR2} mice in a different order, i.e., first PLNx to disrupt the bladder nerve and then PDNx to additionally disrupt the urethral sphincter nerve (**Figure 5A**; see Methods for detail). Interestingly, under the PLNx condition with a filled bladder, PMC^{ESR1+} cell photoactivation did not induce a bladder pressure upstroke (Δ pressure: 3.2\0.96-4.9 cm H₂O in pre-transection control condition, -1.2\1.4-0.8 cm H₂O in PLNx condition, n = 8 mice, $p = 7.8e-3$, Wilcoxon signed-rank test), but the EUS-EMG activity remained responsive, leading to urine leakage which consistently reduced bladder pressure (**Figure 5B-E**). Both the inversed bladder response and the EUS-EMG response were then fully abolished after the second transection, PDNx (**Figure 5B-E-figure supplement 8**). Consistently, under the PLNx condition with an unfilled bladder, the EUS-EMG response was present, and the bladder pressure did not change upon PMC^{ESR1+} cell photoactivation (**Figure supplement 8**). Collectively, these data together indicate that PMC^{ESR1+} cell control of the urethra sphincter was operational even when the pelvic nerve to the bladder was severed.

Lastly, despite that the PMC^{ESR1+} cell photoactivation consistently elicited EUS-EMG bursting in 100% of cases, a refined analysis revealed that the key parameters of the sphincter were slightly degraded under PLNx condition (EMG bursting AUC area: 0.21\0.20-0.22 mV*s in pre-transection, 0.17\0.15-0.19 mV*s in PLNx; EMG bursting duration: 4.9\4.8-5.03 s in pre-transection, 4.5\4.1-4.6 s in PLNx, n = 8 mice, $p = 7.8e-3$; Wilcoxon signed-rank test; **Figure 5E**), which implies a potential reduction in effective voiding volume (Langdale and Grill, 2016). To further explore this, we conducted experiments on awake, unrestrained mice under the PLNx condition (**Figure 6A**; see Methods for detail). Compared to the pre-transection control condition (‘baseline’), PMC^{ESR1+} cell

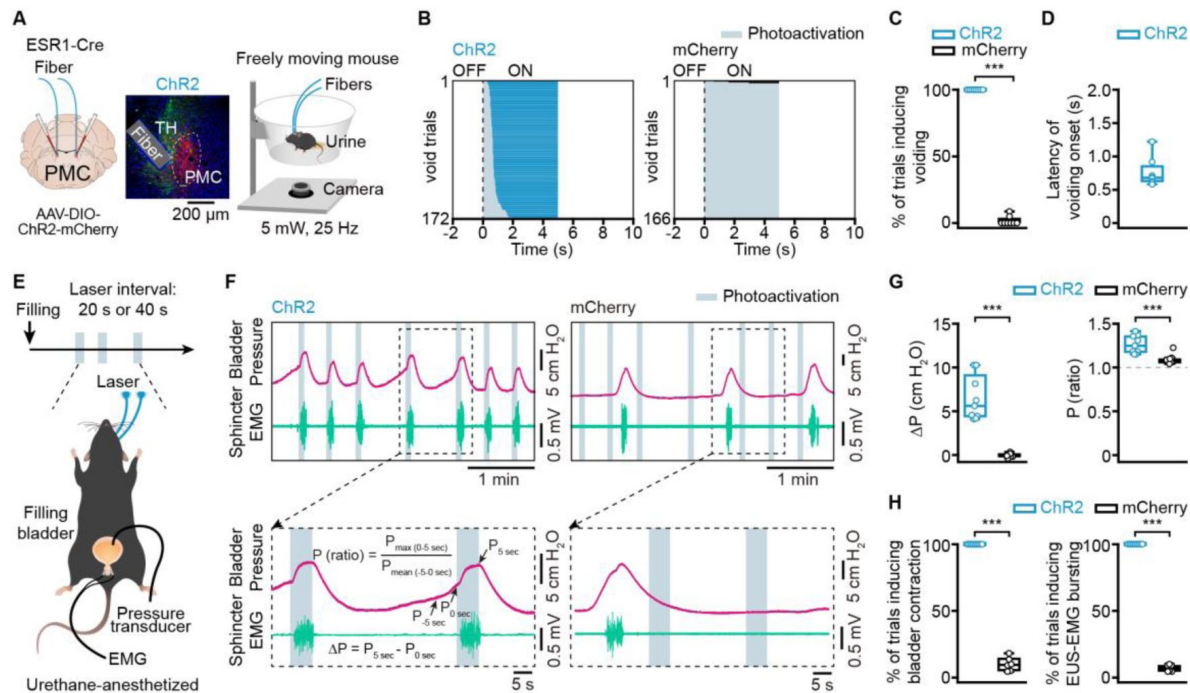


Figure 3.

Activation of PMC^{ESR1+} cells induces both bladder contraction and sphincter relaxation to initiate voiding

(A) Schematics of labeling (left), representative histology (middle), and behavior test (right) for PMC^{ESR1+} photoactivation. Scale bar: 200 μ m. (B) Cumulative trials of voiding duration in PMC^{ESR1-ChR2} (blue bar, n = 172 trials from 8 mice) and PMC^{ESR1-mCherry} (black bar, n = 166 trials from 8 mice) photoactivation. Voiding trials are ordered by the latency of the voiding epoch with the laser on. (C) % of photoactivation-associated voiding events in PMC^{ESR1-ChR2} and PMC^{ESR1-mCherry} groups (n = 8 mice per group, ***P = 1.6e-4, Wilcoxon rank-sum test). (D) Latency of voiding onset after light on. (E) Timeline (top) and schematics (bottom) for PMC^{ESR1+} photoactivation during simultaneous cystometry and electromyography recording. (F) Representative traces (top) and expanded portions (bottom) of bladder pressure (magenta) and EUS-EMG (teal) around photoactivation timepoint in PMC^{ESR1-ChR2} (left) or PMC^{ESR1-mCherry} (right) groups. (G, H) Quantification of bladder pressure change (ΔP , G, left), the ratio of bladder pressure (G, right), the percentage of photoactivation-associated bladder contraction (H, left), and the percentage of photoactivation-associated EUS-EMG bursting (H, right) upon photoactivation (n = 9 PMC^{ESR1-ChR2} mice, n = 6 PMC^{ESR1-mCherry} mice, ***P = 4e-4 for (G, H), Wilcoxon rank-sum test). For all data points in (C, D, G), and (H), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

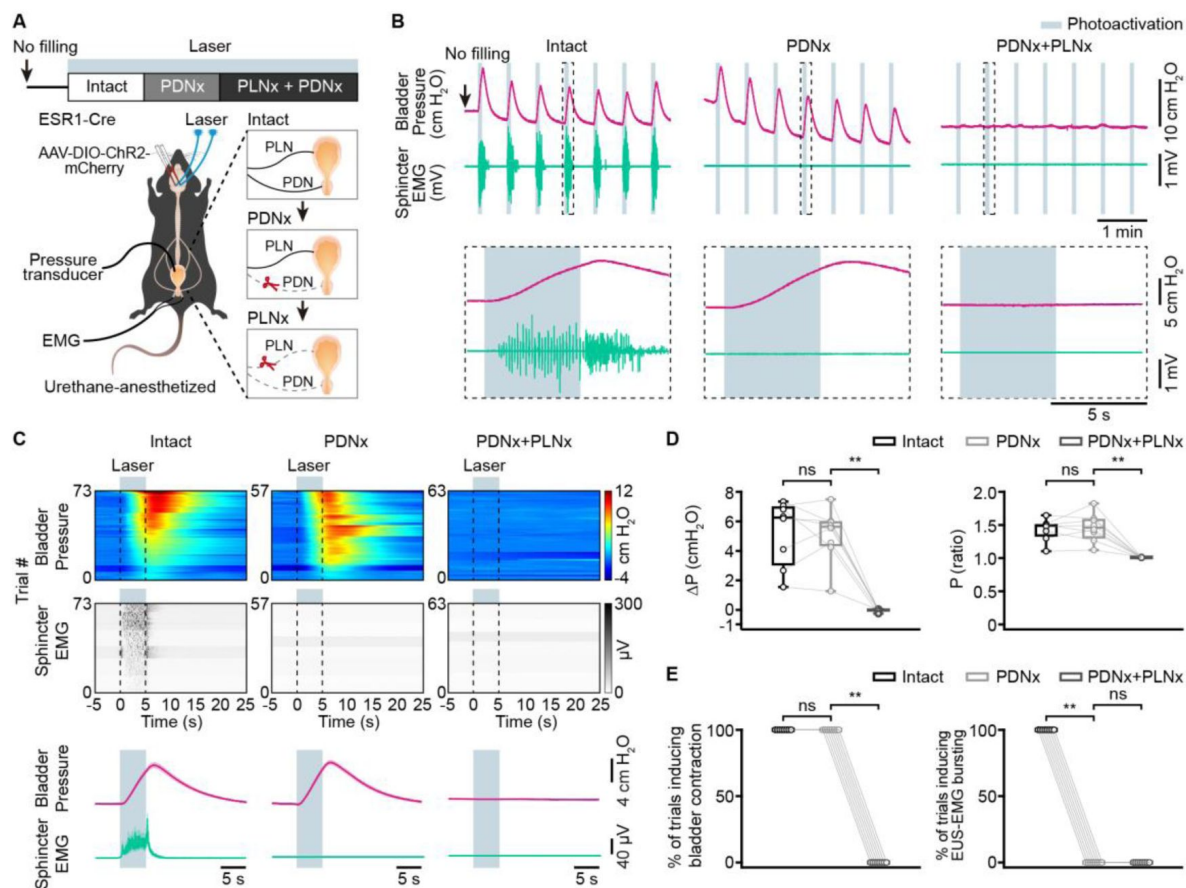


Figure 4.

Transection of the pudendal nerves does not impair bladder contraction induced by PMC^{ESR1+} cell photoactivation.

(A) Timeline (top) and schematic (bottom) for PMC^{ESR1-ChR2} photoactivation during simultaneous cystometry and urethral electromyography recordings, with PDNx performed first. PDNx, pudendal nerve transection; PLNx, pelvic nerve transection. (B) Representative traces (top) and expanded portions (bottom, from the dashed box in the top panel) showing bladder pressure (magenta) and EUS-EMG (teal) during PMC^{ESR1-ChR2} photoactivation in various groups, with PDNx performed first. (C) Heatmap (top) and average traces (bottom; thick lines and shading represent mean $\pm$ s.e.m.) of sorted bladder pressure and EUS-EMG around the photoactivation timepoint for all unfilled bladder trials with the PDNx-first experiment ($n = 8$ mice per group). (D, E) Quantification of bladder pressure change (ΔP , **D**, left), bladder pressure ratio (**D**, right), the percentage of photoactivation-associated bladder contraction (**E**, left), and the percentage of photoactivation-associated EUS-EMG bursting (**E**, right) upon photoactivation for the PDNx-first experiment from (**C**) ($n = 8$ mice per group, from left to right, **D**: $P = 0.8$, $**P = 7.8 \times 10^{-3}$, $P = 0.7$, $**P = 7.8 \times 10^{-3}$, respectively; **E**: $P = 1$, $**P = 7.8 \times 10^{-3}$, $**P = 7.8 \times 10^{-3}$, $P = 1$, respectively; n.s., not significant; Wilcoxon signed-rank test). For all data points in (**D**, **E**), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

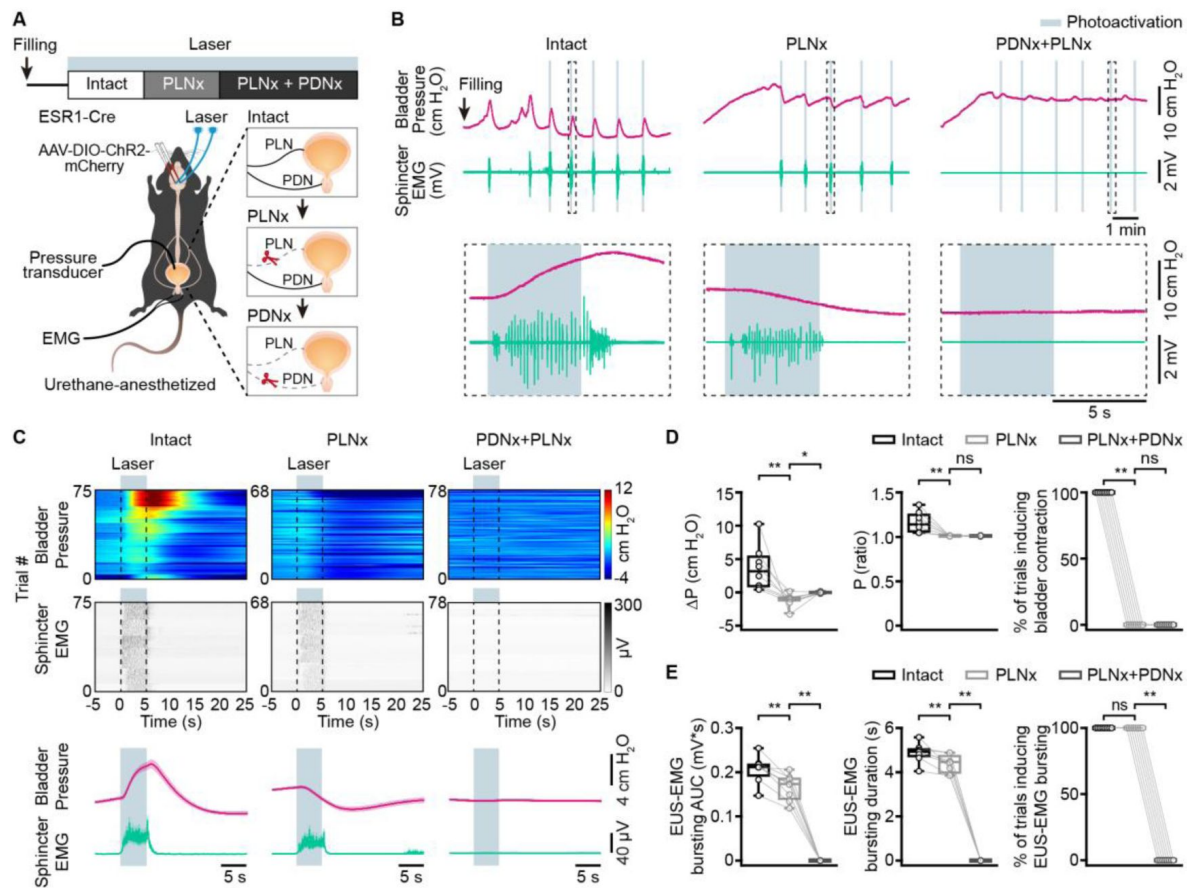


Figure 5.

Transection of the pelvic nerves does not impair sphincter relaxation induced by PMC^{ESR1+} cell photoactivation.

(A) Timeline (top) and schematic (bottom) for PMC^{ESR1-ChR2} photoactivation during simultaneous cystometry and urethral electromyography recordings, with PLNx performed first. PLNx, pelvic nerve transection; PDNx, pudendal nerve transection. (B) Representative traces (top) and expanded portions (bottom, from the dashed box in the top panel) showing bladder pressure (magenta) and EUS-EMG (teal) during PMC^{ESR1-ChR2} photoactivation in various groups, with PLNx performed first. (C) Heatmap (top) and average traces (bottom; thick lines and shading represent mean $\pm$ s.e.m.) of sorted bladder pressure and EUS-EMG around photoactivation timepoint for all filled bladder trials with the PLNx-first experiment ($n = 8$ mice per group). (D) Quantification of bladder pressure parameters for the PLNx-first experiment from **c**: bladder pressure change (ΔP , left), bladder pressure ratio (middle), and the percentage of photoactivation-associated bladder contraction (right; from left to right: $**P = 7.8e-3$, $P = 0.02$, $**P = 7.8e-3$, $P = 0.6$, $**P = 7.8e-3$, $P = 1$, respectively; n.s., not significant; Wilcoxon signed-rank test). (E) Quantification of EUS-EMG parameters for the PLNx-first experiment from **c**: EUS-EMG bursting AUC (left), EUS-EMG bursting duration (middle), and the percentage of photoactivation-associated EUS-EMG bursting (right; from left to right: $**P = 7.8e-3$, $**P = 7.8e-3$, $**P = 7.8e-3$, $**P = 7.8e-3$, $P = 1$, $**P = 7.8e-3$, respectively; n.s., not significant; Wilcoxon signed-rank test). For all data points in **(D, E)**, whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

photoactivation still triggered voiding in 100% of trials (**Figure 6B** and **C**). However, in PLN $\times$ mice, the voiding events were characterized by significantly smaller urine spots and a longer latency to voiding onset than the control events before transection, a difference not observed in the sham surgery group (**Figure 6D** and **E**). These findings could be interpreted as that while the PMC^{ESR1+}-EUS pathway is sufficient to drive voiding, as previously reported (Keller et al., 2018), however, without an intact bladder reflex pathway involved, an efficient urine flow cannot be performed. Piecing these data of the combined transection-optogenetics experiments together, we reveal a more complete picture that PMC^{ESR1+} cells can operate the bladder (through the pelvic nerve) and the sphincter (through the pudendal nerve) independently of each other. Such refined knowledge could not have been obtained from simpler tests using optogenetics alone as shown above (**Figures 2** and **3**), or from other experiments in the literature (Hou et al., 2016; Keller et al., 2018; Ito et al., 2020).

PMC^{ESR1+} cells coordinate bladder contraction and sphincter relaxation to initiate urination

The above data showing that PMC^{ESR1+} cells can operate through both the bladder and urethral sphincter independently of each other implies that there should be two distinct anatomical projections from PMC^{ESR1+} cells downstream to innervate these targets. Indeed, our anterograde labeling experiment in ESR1-Cre mice (**Figure supplement 9**; see Methods for detail) showed that the mGFP-labelled axonal terminals of the PMC^{ESR1+} cell population were found in both the DGC and the SPN region of the lumbosacral spinal cord, which further send axons via the pudendal and pelvic nerves to innervate urethral sphincter and bladder, respectively. To further test this hypothesis, we performed retrograde labeling experiments in ESR1-Cre mice and, as a control, in CRH-Cre mice (**Figure 7A** and **E**; see Methods for detail). Analysis of CRH-Cre mice showed that 82.4% (80.1%–86.7%) of PMC^{CRH+} cells were SPN-projecting only (expressing mCherry), and 1.8% (1.4%–2.2%) were DGC-projecting only (expressing EGFP), whereas the remaining 15.8% (16.8%–19.1%) were dual-projecting (altogether $n = 1416$ cells pooled from 79 slices obtained from 3 mice; **Figure 7B–D**). This result is consistent with the literature (Valentino et al., 2010; Ito et al., 2020; Hou et al., 2016; Keller et al., 2018) that PMC^{CRH+} cells primarily project to the SPN in the spinal cord and modulate bladder contraction. By contrast, analysis of the ESR1-Cre mice showed that 19.2% (24.1%–24.2%) of PMC^{ESR1+} cells were SPN-projecting only, 53.3% (50.9%–55.2%) were DGC-projecting only, and 27.5% (26.8%–30.9%) cells were dual-projecting ($n = 1417$ cells pooled from 82 slices obtained from 5 mice; **Figure 7F–H**). These data suggest that PMC^{ESR1+} cells innervate either SPN or DGC in a more balanced manner and include a significant fraction of dual-innervating cells, in contrast to PMC^{CRH+} cells that primarily innervate the SPN.

The functional and anatomical data above together suggest that PMC^{ESR1+} cells can function as a 100% reliable ‘master switch’ either to initiate or to suspend voiding through independently operating the bladder (via SPN to the pelvic nerve) and the sphincter (via DGC to the pudendal nerve). We now come to the final question as to whether PMC^{ESR1+} cells can implement the coordination between the bladder and urethra sphincter, i.e., operate them in a rigid temporal order. We took advantage of the simultaneous recording of cystometry and EUS-EMG condition (e.g. **Figure 2**) in which the beginning/ending timepoint of bladder pressure upstroke (a significant rapid, transient increase in bladder pressure preceding voiding, denoting threshold pressure of bladder contraction) (Rana et al., 2024) and the stereotypic voiding-associated firing pattern of EUS-EMG (denoting sphincter relaxation) could be precisely determined. In the photometry recording experiments when voiding events spontaneously occurred, the bladder pressure upstroke timepoint always preceded the beginning timepoint of the EUS-EMG firing pattern (bladder pressure upstroke onset: 0.6–1.2 s, EMG bursting onset: 2.3–3.2 s, relative to the onset of Ca²⁺ signals; $n = 46$ events pooled from 8 mice; **Figure 8A**). Accordingly, in the photoactivation experiments, the bladder pressure upstroke timepoint also preceded the EMG

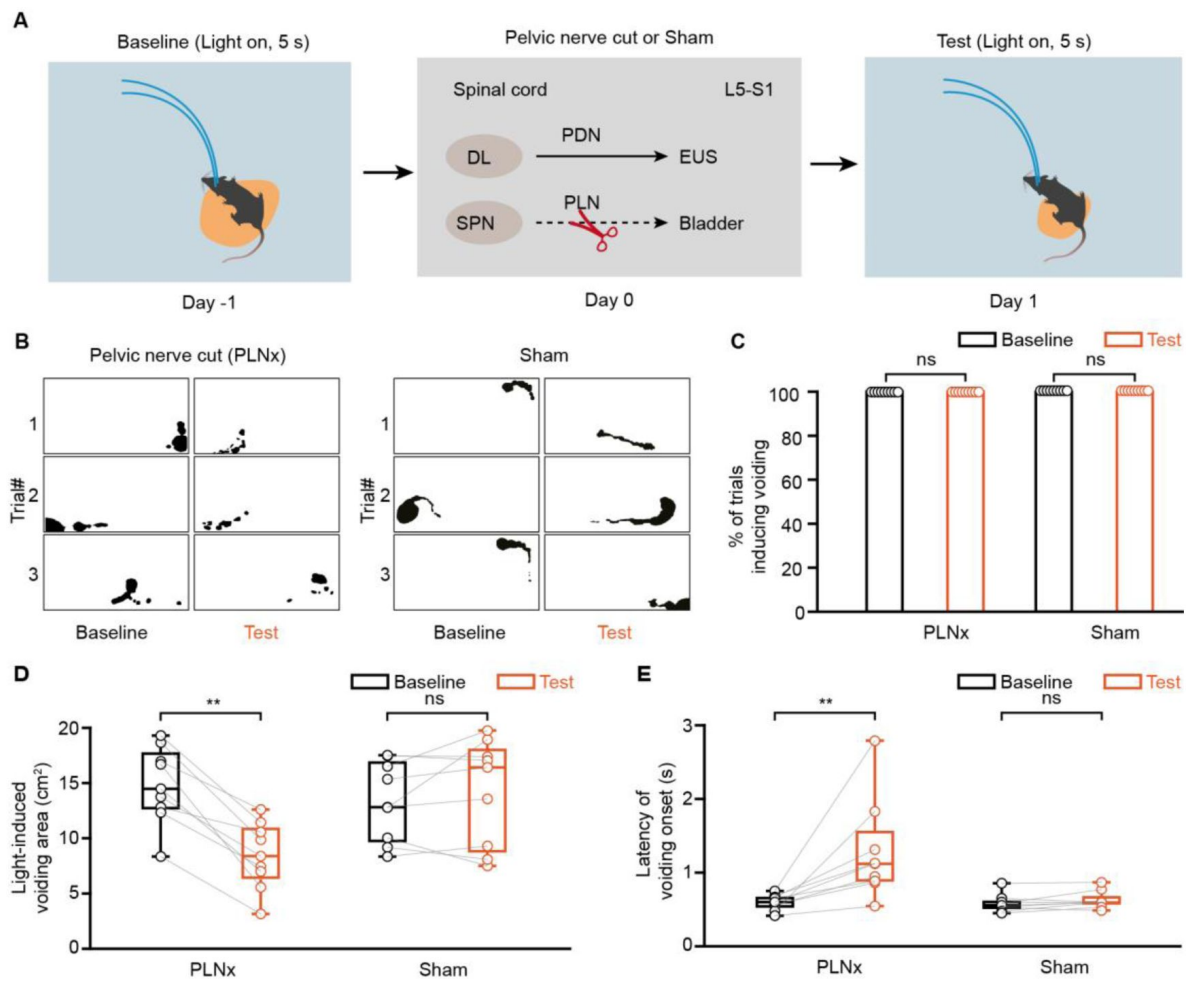


Figure 6.

Transection of the pelvic nerves decreases urinary volume induced by PMC^{ESR1+} cell photoactivation.

(A) Schematic (top) and timeline (bottom) for PMC^{ESR1-ChR2} photoactivation during simultaneous cystometry and urethral electromyography recordings in a freely moving mouse with pelvic nerve transection (PLNx). (B) Representative images of light-induced urination marking (black shading) in ESR1-Cre mice before ('Baseline') and after ('Test') pelvic nerve transection (left) or sham transection (right). (C-E) Quantification of the effect of pelvic nerve transection on voiding in the PLNx (n = 9 mice) and sham groups (n = 9 mice): the percentage of light-induced voiding (C, $P = 1$; n.s., not significant; Wilcoxon signed-rank test), light-induced voiding area (D, $**P = 3.9 \times 10^{-3}$, $P = 0.5$, respectively), and latency of voiding onset after light stimulation (E, $**P = 3.9 \times 10^{-3}$, $P = 0.2$, respectively; n.s., not significant; Wilcoxon signed-rank test). For all data points in (D, E), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

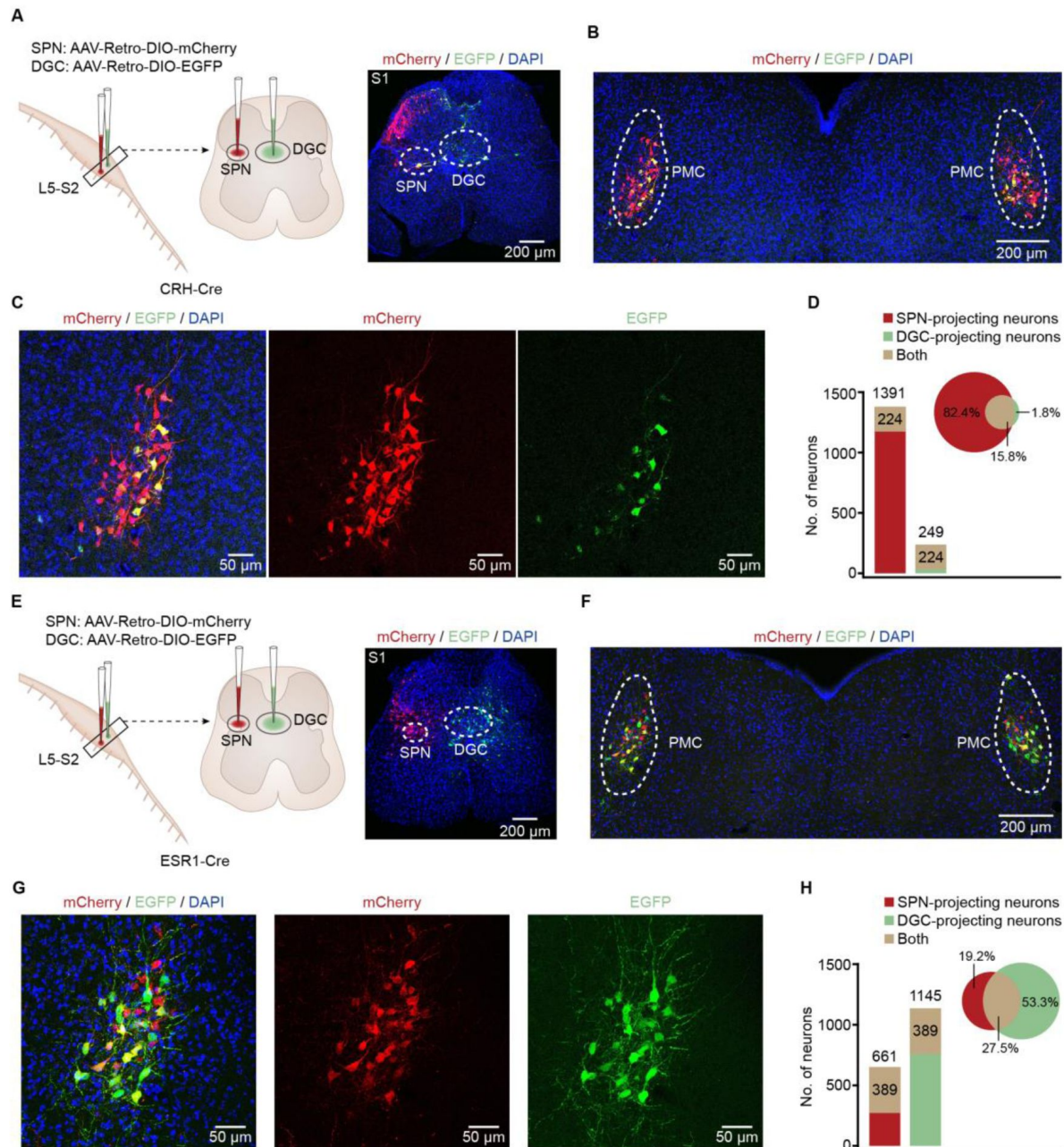


Figure 7.

Differences in the anatomical projections to the lumbosacral spinal cord between PMC^{CRH+} and PMC^{ESR1+} cells.

(A) Schematic of labeling (left) and representation histology of virus expression (right) at the spinal cord of CRH-Cre mice. Scale bar: 200 μ m. (B, C) Representative image (B) and enlarged images (C, from the left part of B) showing EGFP and mCherry expression in the PMC of a CRH-Cre mouse. Scale bars: 200 μ m for (B) and 50 μ m for (C). (D) Quantification of the fractions of CRH⁺ cells specifically projecting to the SPN and DGC of the spinal cord, respectively (n = 1416 cells from 3 mice). (E) Schematics of labeling (left) and representative histology of virus expression (right) in the spinal cord of ESR1-Cre mice. Scale bar: 200 μ m. (F, G) Representative image (F) and enlarged images (G, from the left part of F) showing EGFP and mCherry expression in the PMC of an ESR1-Cre mouse. Scale bars: 200 μ m for (F) and 50 μ m for (G). (H) Quantification of the fractions of ESR1⁺ cells specifically projecting to the SPN and DGC of the spinal cord, respectively (n = 1417 cells from 5 mice). SPN, sacral parasympathetic nucleus; DGC, dorsal gray commissure.

bursting pattern begin timepoint, albeit both were slightly earlier than those in ‘passive’ spontaneous photometry recordings (bladder pressure upstroke onset: 0.4\0.4-0.5 s, EMG bursting onset: 0.8\0.7-1.0 s; $n = 50$ events pooled from 10 mice; **Figure 8B** [↗](#)). Despite that the photoactivation of the $\text{PMC}^{\text{ESR1}+}$ cells could have been artificially strong and did not necessarily mimic the naturalistic firing pattern of $\text{PMC}^{\text{ESR1}+}$ cells (see **Figure 1G-I** [↗](#)), the same temporal order as bladder contraction preceded sphincter relaxation suggests that the downstream circuitry was instructed to execute in the same temporal order. Intriguingly, in photoinhibition experiments, the EMG pattern almost instantaneously ended, while the bladder pressure upstroke end timepoint occurred later (EMG bursting end: 0.1\0.1-0.2 s; bladder pressure upstroke end: 0.8\0.8-0.9 s, $n = 44$ events pooled from 8 mice; **Figure 8C** [↗](#)). These timing results can be interpreted from a broader perspective by considering the lower spinal reflex circuit together with $\text{PMC}^{\text{ESR1}+}$ projections from the brainstem (**Figure 8D** [↗](#)). Since the lower spinal reflex circuit is the forefront agent that can send a feedforward reflex signal from the bladder to the sphincter unidirectionally (Lee et al., 2021 [↗](#); de Groat et al., 2015 [↗](#); Chang et al., 2007 [↗](#); Abud et al., 2015 [↗](#)), the brainstem coordination signal descending in two parallel paths from $\text{PMC}^{\text{ESR1}+}$ cells would result in the initiation of voiding with the same forwarding order but in the suspension of ongoing voiding with the reversed order.

Discussion

A prominent aspect of our data is that voiding events 100% correlated with $\text{PMC}^{\text{ESR1}+}$ neuronal activity (**Figure 1** [↗](#)), and both photoactivation and photoinhibition of $\text{PMC}^{\text{ESR1}+}$ cells yielded a 100% success rate in initiating and suspending the process of voiding, respectively (**Figure 2** [↗](#) and **3** [↗](#)). Whilst there are many other factors from both the lower spinal circuit and the higher cortical/subcortical circuits together to determine in what condition to void (Yao et al., 2018 [↗](#); de Groat et al., 2015 [↗](#); Sartori et al., 2022 [↗](#); de Groat, 2009 [↗](#); Zderic, 2019 [↗](#); Mukhopadhyay and Stowers, 2020 [↗](#)), the instantaneous execution to efficiently initiate or suspend a voiding process involves a significant contribution of $\text{PMC}^{\text{ESR1}+}$ cells in the brainstem as demonstrated here in this study. The bidirectional control, i.e., in the one way to initiate voiding whenever needed and suitable (which is a basic life need) (Mukhopadhyay and Stowers, 2020 [↗](#)), and in the other way to suspend an ongoing voiding when needed (e.g., to release only a small volume of urine for landmarking) (Keller et al., 2018 [↗](#); Hyun et al., 2021 [↗](#); Desjardins et al., 1973 [↗](#)), can be executed at 100% reliability by $\text{PMC}^{\text{ESR1}+}$ cells given that all downstream nerves and muscles are intact and operational.

Moreover, the anatomical data also revealed a significant fraction of dual-projecting $\text{PMC}^{\text{ESR1}+}$ cells (**Figure 7H** [↗](#)) which we consider to play a significant role in the coordination of the bladder and sphincter. Because these two projections originate from the same cells, the ‘on’ and ‘off’ commands for bladder contraction sent from these cells would always align with the ‘on’ and ‘off’ commands for sphincter relaxation, respectively. As such, the brainstem can serve the upstream higher brain regions (Yao et al., 2018 [↗](#); Zare et al., 2019 [↗](#); Manohar et al., 2017 [↗](#)) (e.g., the neocortex) with a 100% reliable ‘master switch’ to urinate a desired volume whenever in physiological or social needs (Mukhopadhyay and Stowers, 2020 [↗](#); Malykhina, 2017 [↗](#); Kaur et al., 2014 [↗](#)), e.g., to mark territories with small urine spots as an animal, or to urinate into a small sample tube for a health check as a human.

The fact that $\text{PMC}^{\text{ESR1}+}$ cells can execute dynamic, real-time control of voiding at 100% reliability does not imply that other cell types in the PMC are not required to achieve the same goal of perfect urination control. On the contrary, we suggest that a proper baseline control of bladder pressure is also mandatory, e.g., by CRH^+ cells which are the majority of cells in PMC that primarily operate the bladder through the pelvic nerve and can be modulated by various contextual factors (Hou et al., 2016 [↗](#); Vincent and Satoh, 1984 [↗](#); Wood et al., 2009 [↗](#)). After all, it is not the cell type domain marker ESR1 (which by itself abundantly expressed in many other regions of the brain and is

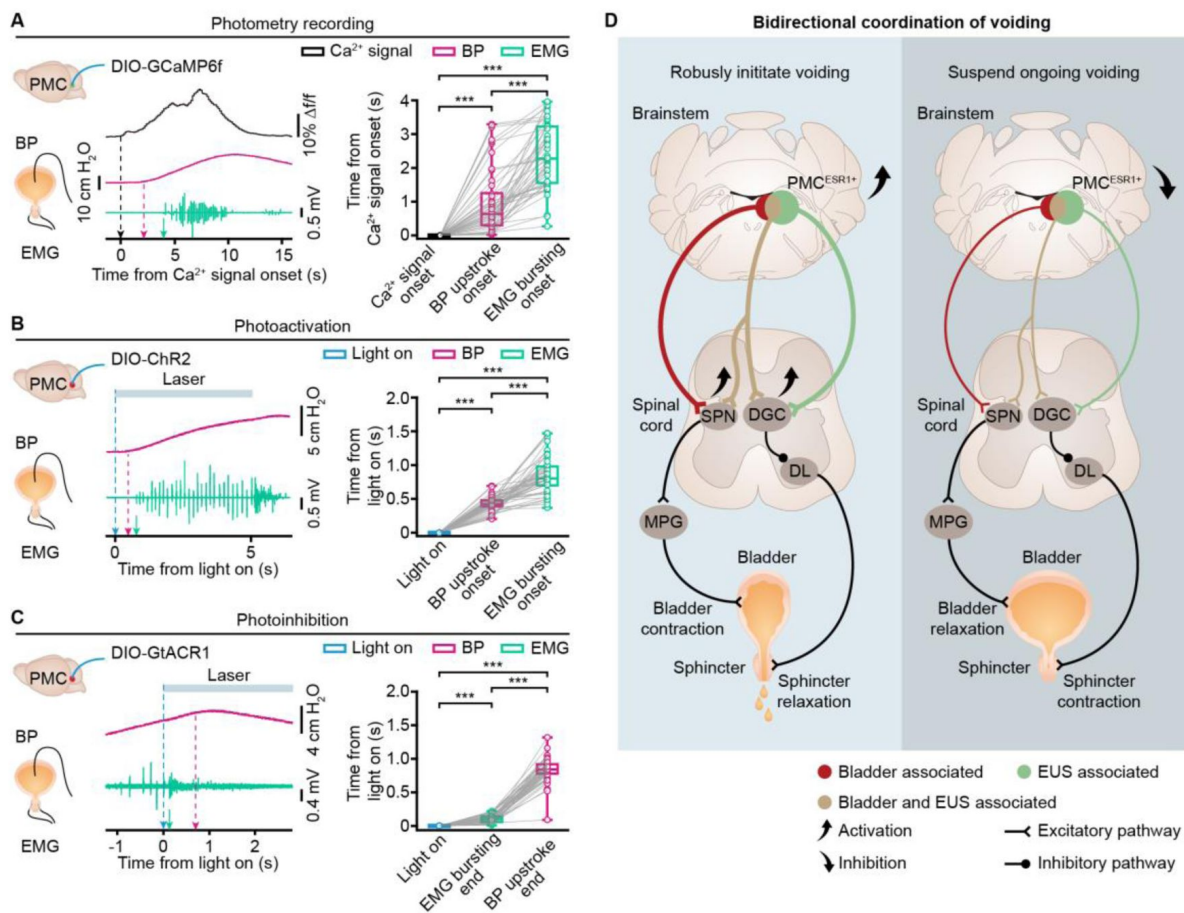


Figure 8.

Coordination of bladder contraction and sphincter relaxation for urination by PMC^{ESR1+} cells.

(A) Left: Example (left, with arrows indicating the onset timepoints) and quantification (right) of the temporal relationships among the onset times of Ca^{2+} signals, bladder pressure (BP) upstroke, and EUS-EMG bursting in photometry recordings ($n = 46$ trials from 8 mice; Ca^{2+} signals onset, 0 ± 0 s; BP upstroke onset, 0.6 ± 0.3 – 1.2 s; EUS-EMG bursting onset, 2.3 ± 1.5 – 3.2 s; $***P = 3.5 \times 10^{-9}$ for all, Wilcoxon signed-rank test). (B) Example (left, with arrows indicating the onset timepoints) and quantification (right) of the temporal relationships among the onset times of light stimulation, bladder pressure upstroke, and EUS-EMG bursting for the photoactivation group ($n = 50$ trials from 10 mice; Light on, 0 ± 0 s; BP upstroke onset, 0.4 ± 0.4 – 0.5 s; EUS-EMG bursting onset, 0.8 ± 0.7 – 1.0 s; $***P = 7.6 \times 10^{-10}$ for all, Wilcoxon signed-rank test). (C) Example (left, with arrows indicating the onset timepoints) and quantification (right) of the temporal relationships among the onset time of light, EUS-EMG bursting end, and bladder pressure upstroke end for the photoinhibition group ($n = 44$ trials from 8 mice; Light on, 0 ± 0 s; EUS-EMG bursting end, 0.1 ± 0.1 – 0.2 s; BP upstroke end, 0.8 ± 0.8 – 0.9 s; $***P = 7.6 \times 10^{-9}$ for all, Wilcoxon signed-rank test). (D) A working model of the brainstem-spinal cord circuit for bidirectional and coordinated control of voiding. Abbreviations: SPN, sacral parasympathetic nucleus; DGC, dorsal gray commissure; DL, dorsolateral nucleus; MPG, major pelvic ganglia. For all data points in (A, B), and (C), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

involved in many other physiological and cognitive functions (Karigo et al., 2021 [↗](#); Liu et al., 2022 [↗](#); Fang et al., 2018 [↗](#)), but rather the topology of innervation (**Figure 8D** [↗](#)) that directly enables such a role of urination coordination, i.e., a cell of whichever marker that locates in this particular brainstem nucleus (PMC) (Kawatani et al., 2021 [↗](#)) and possesses dual innervations towards both the bladder and sphincter is a potent contributor to urination coordination (Griffiths, 2015 [↗](#)). Nevertheless, our pinpointing of PMC^{ESR1+} cells performing a 100% reliable role in bladder-sphincter coordination enlightens the future development of precise therapies for treating urination coordination disorders.

Conclusions

In summary, we have identified the essential role of PMC^{ESR1} neurons in coordinating urination through co-innervation of both the bladder and urethra sphincter. Our findings offer new insights into the anatomical and physiological basis and research paradigms for the coordinated engagement of parasympathetic and somatic functions of urination control. PMC^{ESR1} neurons may serve as a key focal point for advancing our understanding of the neural mechanisms underlying urination, both in physiological contexts and pathological conditions such as brain injuries, spinal cord injuries, and peripheral nerve damages.

Materials and methods

Animals

The experiment procedures were approved by the Third Military Medical University Animal Care and Use Committee and were conducted strictly in adherence to established guidelines. This study utilized ESR1-IRES-Cre (Jackson Laboratory, stock #017911) (Lee et al., 2014 [↗](#)) and CRH-IRES-Cre (Jackson Laboratory, stock #012704) (Baram et al., 2015 [↗](#)) mice. These mice were group-housed in an environment-controlled room at 23-25°C and 50% humidity, with 4-5 mice per cage, on a 12-hour light/dark cycle, and with free access to food and water. Mice implanted with optical fibers were housed individually. Both male and female mice, aged 8-20 weeks, were randomly assigned to various experiments. The figure legends and supplement tables detail the number of animals utilized in each experiment.

Virus vectors and CTB

The study utilized the following viruses and cholera toxin subunit B (CTB) for various experiments: For fiber photometry recording experiments, AAV2/9-DIO-GCaMP6f (titer: 0.5×10^{12} vg/ml) was unilaterally injected into the PMC. For optogenetic manipulation experiments, AAV2/9-hsyn-DIO-hGtACR1-mCherry (titer: 1.43×10^{13} vg/ml) and AAV2/8-DIO-ChR2-mCherry (titer: 1.33×10^{13} vg/ml) were bilaterally delivered into the PMC for photoinhibition and photoactivation, respectively. For retrograde tracing experiments, pAAV2/retro-EF1a-DIO-EGFP (titer: $\geq 1.00 \times 10^{12}$ vg/ml) and pAAV2/retro-EF1a-DIO-mCherry (titer: $\geq 1.00 \times 10^{12}$ vg/ml) were injected into the spinal cords of ESR1-Cre and CRH-Cre mice, respectively. For anterograde tracing experiments, AAV2/9-hEF1a-fDIO-mGFP (titer: 1.00×10^{13} vg/ml) was injected unilaterally into the PMC, and a 10: 1 volume mixture of AAV2/2Retro-hsyn-FLEX-Flpo (titer: 1.00×10^{13} vg/ml) and CTB 555 (0.2%, C34776, Thermo Fisher; used solely for determining the injection site) was injected into the spinal cord. In control experiments, rAAV2/9-EF1a-DIO-EYFP (titer: 4.20×10^{12} vg/ml) and rAAV2/9-EF1a-DIO-mCherry (titer: 5.28×10^{12} vg/ml) were used. All viruses were purchased from Obio Biotechnology Co., Ltd. (Shanghai, China), Taitool Bioscience Co., Ltd. (Shanghai, China), or BrainVTA Co., Ltd. (Wuhan, China).

Stereotaxic injections and optical fiber implant

To target PMC^{ESR1+} neurons, ESR1-Cre mice were anesthetized with isoflurane (3% for induction and 1.5–2% for maintenance) and head-fixed in a stereotaxic frame (RWD Life Science Co., Ltd.; Shenzhen, China). Body temperature was maintained at 36°C throughout the surgery using a heating pad. Local anesthesia (lidocaine, 6 mg/kg, subcutaneous injection) was administered at the incision site before making the incision. A small cranial hole above the PMC was created using a dental drill. Approximately 80 nl of the viral solution was delivered to the injection site at a controlled rate of 20 nl/min, either unilaterally or bilaterally, using a micro-syringe pump connected to a glass pipette. The coordinates for PMC injections were: anteroposterior (AP) is -5.45 mm, mediolateral (ML) is $\pm$ 0.70 mm, and dorsoventral (DV) is -3.14 mm from the dura. The pipette was then slowly withdrawn over 5 min to prevent virus overflow. The incision was closed with sutures, and the mice received antibiotics and analgesics post-surgery. A heating pad was used for the mice to aid recovery from anesthesia. The mice were then group-housed in their home cages for 3–4 weeks to allow for viral expression.

For optic fiber implantation, the optic fiber (NA: 0.48, diameter: 200 μ m, Doric lenses, Quebec City, QC, Canada) was fixed into a metal cannula and positioned 50 μ m above the PMC injection site. The fiber was implanted unilaterally for photometry experiments and bilaterally for optogenetics experiments. Dental cement was used to affix the optical fiber to the skull. Mice were housed individually and given 3–5 days to recover before recording or stimulation sessions. After the experiments, the placement of the virus and optical fiber was confirmed by histology in each mouse.

Targeted spinal cord injections

For spinal cord injection surgery, the method previously described was used (Chen et al., 2019 [link](#)). Briefly, mice were anesthetized under isoflurane (1.5–2% oxygen) and positioned on a heating pad. After shaving the hair, a midline skin incision (1–2 cm) was performed over the lumbar segments following local anesthesia (lidocaine, 6 mg/kg, subcutaneous injection) at the incision site. The tissue and muscle connected to the dorsal spine were dissected to expose the T12–L2 vertebrae. The spine was affixed to a stereotactic frame using a spinal adapter (68094, RWD), and the spinal cord between L1 and L2 was exposed by removing the ligamentous and epidural membranes. Using the central vein as a reference, a total of 80 nL of virus solution was injected at two different locations, each injection (40 nL) was administered through a micro-syringe pump connected to a glass pipette, targeting DGC (+ 0.12 mm lateral to the central vein, 0.52 mm deep from the dorsal surface at a 10-degrees angle) and SPNs (- 0.12 mm lateral to the central vein, 0.52 mm deep from the dorsal surface at a 30-degrees angle). The pipette remained in position for a minimum of 5 min before being slowly removed to prevent any leakage. The injection sites were sealed with tissue glue (Vetbond, 3M Animal Care Products), followed by suturing of the skin. Mice received antibiotics and analgesics post-surgery. Approximately 3–4 weeks after the injections, the brain and spinal cords were extracted for histological validation. In ESR1-Cre or CRH-Cre mice, PMC cells projecting to the SPN (labeled with mCherry) and DGC (labeled with EGFP) regions of the spinal cord were manually quantified using Image J.

Pudendal nerve and pelvic nerve transection

Pudendal nerve transection was performed as described previously (Khorramirouz et al., 2016 [link](#); Peh et al., 2018 [link](#)). Briefly, mice were anesthetized with isoflurane (1.5–2% oxygen) and positioned on a heating pad. A midline skin incision was made along the back from L4 to the coccyx, followed by paraspinal incisions through the gluteal muscles and fascia to expose the sciatic nerve. The sciatic nerve was gently retracted to expose the pudendal nerve. Using microsurgical scissors, the bilateral pudendal nerves, along with the anastomotic branch, were carefully dissected and excised.

For pelvic nerve transection, modifications to the established method (Chang et al., 2018) were made to minimize surgical trauma. Mice were positioned laterally, and bilateral paraspinal incisions were extended upward. The sciatic and pudendal nerves were gently retracted laterally to expose the pelvic nerve, which was identified (originating from the sacral segments of the spinal cord and connecting to the major pelvic ganglion) and severed. For pelvic nerve transection experiments in freely moving mice, the muscles and skin were sutured, and antibiotics and analgesics were administered. In the sham injury group, the same procedures were followed, except for the transection of the pelvic nerve.

Fiber photometry recording and analysis in freely behaving mice

The Ca^{2+} recordings of $\text{PMC}^{\text{ESR1}+}$ neurons were conducted using a fiber photometry setup, as described previously (Yao et al., 2018; Rao et al., 2022). Fluorescence at the fiber tip was excited by blue light (0.22 mW/mm^2). Mice with implanted fibers were injected intraperitoneally with diuretics (furosemide, 40 mg/kg) and acclimated in a testing chamber equipped with a bottom camera ($1,280 \times 720$ pixels) for 20 min before recording. Signals from $\text{PMC}^{\text{ESR1}+}$ neurons and voiding behavior were simultaneously recorded for approximately 40 min. Ca^{2+} signals were sampled at 2000 Hz using NI LabVIEW software (National Instruments, USA), while behavioral video was captured at 30 Hz. Fiber photometry data and video were synchronized via event markers. For data analysis, all signals were processed with a Savitzky-Golay filter (third-order polynomial, 50 side points) for low-pass filtering. $\Delta f/f = (f - f_{\text{baseline}})/f_{\text{baseline}}$ was calculated to assess photometry signals during voiding, where f_{baseline} represents the minimum fluorescence recorded. Results were presented as heatmaps using MATLAB. Ca^{2+} signal data were shuffled by dividing the original dataset into 10 segments and randomly associating them with voiding events. Positive signals were defined as Ca^{2+} signal amplitudes exceeding three times the noise band (the standard deviation). This procedure was also applied to control mice to correct for movement artifacts.

Single-unit with optrode recording and analysis

To identify the single-unit activity of $\text{PMC}^{\text{ESR1}+}$ neurons, optrode recordings were performed as described previously (Qin et al., 2022; Qin et al., 2018; Yang et al., 2023). Briefly, the optrode consisted of a $200 \mu\text{m}$ optical fiber and four tetrode assemblies aligned in a line, spaced $100 \mu\text{m}$ apart. The optical fiber was secured to the electrodes, positioned $500 \mu\text{m}$ above their tips, and connected to an LED. Each electrode assembly comprised four twisted tungsten wires ($25 \mu\text{m}$, California Fine Wire), allowing vertical movement via micromanipulators. Optrode implantation surgery was performed in ESR1-Cre mice expressing ChR2 in the PMC, with the electrode tips aligned and implanted 2.80 mm below the brain dura. After a recovery period of 5-7 days, the tetrodes were slowly inserted to a target depth of -2.95 mm and recording began. Single-unit signals from $\text{PMC}^{\text{ESR1}+}$ neurons in freely behaving mice were recorded using an RHD2000 USB board (C3100, Intan Technology) at 20 kHz , while behavioral video was captured simultaneously. Units with short spike latencies ($< 7 \text{ ms}$) in response to light pulses of varying intensities (5 mW , 10 mW , 15 mW , and 20 mW) and high responsiveness ($> 70 \%$) were identified as $\text{PMC}^{\text{ESR1}+}$ cells. To confirm recording locations, electro-lesions were performed by applying a current ($10 \mu\text{A}$, 12 s) through the tetrodes.

The raw recorded data were preprocessed using established methods to extract peaks (Qin et al., 2018). All events exceeding the amplitude threshold (set at four standard deviations above the background) were kept for further analysis. The average firing rate of each cell was calculated within a sliding time bin of 20 seconds around voiding (0.1 s intervals), divided by the total number of trials, and adjusted by subtracting the baseline value (the median firing rate during the -10 to -5 s before voiding). The results were visualized as a heatmap (logarithmic analysis) in **Figure 1**. Statistical analysis of the average firing rates over a 2-second interval before voiding (from -10 s to -8 s) and around voiding (from -1 s to 1 s) was conducted for $\text{PMC}^{\text{ESR1}+}$ neurons and non- $\text{PMC}^{\text{ESR1}+}$ neurons, respectively.

Optogenetic experiments in freely behaving mice

Before optogenetic stimulation, mice underwent the same procedures as for fiber photometry: they received a diuretic with the intraperitoneal injection to increase urination events and were acclimated to the testing chamber for at least 20 min. For optogenetic inhibition experiments, bilateral stimulation (1 mW/mm^2 at fiber tips, 50 Hz frequency, 20 ms pulses) was delivered using 473 nm blue light. To assess the effect of light inhibition on urination, mice were placed in a glass chamber ($28 \text{ cm} \times 16 \text{ cm} \times 30 \text{ cm}$) with a 0.19 mm filter paper ($14.6 \text{ cm} \times 27 \text{ cm}$, BWD) underneath. Urination was observed at three stages: pre-photoinhibition (light-off), during the 5 s or 60 s of photoinhibition (light-on), and post-photoinhibition (light-off). The photoinhibition parameters (frequency and duration) were controlled via the NI LabVIEW platform (National Instruments, USA). Real-time urination behavior was monitored simultaneously using two cameras positioned above and below the glass chamber (Chongqing NewLight Co., Ltd; see: www.newlightxlr.com). The laser was manually triggered at the onset of urination (within 4 s), and experiments where the trigger exceeded 4 s were excluded. Urination cessation was defined as the initiation of movement after urination. The void area, total void duration, and latency were analyzed from the video and are presented in **Figure 2** and **Figures supplement 5** and **6**.

For optogenetic activation experiments, bilaterally fiber-implanted mice were connected to two 473-nm blue laser generators (5 mW/mm^2 at fiber tips, 25 Hz frequency, 15 ms pulses) via optic fibers. During testing, blue light was delivered for 5 s with approximately 3-min intervals between trials, over a 30-45 min session. This light stimulation was repeated over two days with a one-day interval. Mouse behavior was monitored and recorded simultaneously. The stimulation was performed before and after transection for photoactivation experiments in freely behaving mice with pelvic nerve transection using the same procedures. Note that the photostimulation experiments for pelvic nerve transactions were conducted one day after the surgery. Detailed success rates, latency of voiding, and void area after photoactivation are shown in **Figures 3** and **6**.

Simultaneous cystometry and electromyography in anesthetized mice

Bladder catheter and urethral sphincter electrode implantation were performed as previously described (Verstegen et al., 2019; Hou et al., 2016; Keller et al., 2018). Briefly, adult fiber-implanted ESR1-Cre mice were anesthetized with isoflurane and placed on a heating pad. A lower mid-abdomen incision exposed the bladder and urethral sphincter. PE-10 tubing was inserted through the bladder dome and secured with a 6-0 Ethicon suture. For electromyography (EMG) recording, two 160 μm silver-plated copper wire electrodes (P/N B34-1000, USA) with 2 mm exposed tips were inserted into the external urethral sphincter (EUS) on both sides using a 30-gauge needle, positioned between the urethra and pubic symphysis, and spaced at least 2 mm apart. A ground wire with 4 mm exposed tips was inserted subcutaneously near the sternal notch to minimize signal interference. The ends of the bladder catheter tubing and wire electrodes were exteriorized through an incision on the skin at the back of the neck, and both abdominal and neck incisions were closed. For measuring intravesical pressure, the bladder tubing was connected to a pressure transducer (YPJ01H; Chengdu Instrument Factory, China) and a syringe pump (RWD404; RWD Technology Corp., Ltd., China) via three-way stopcocks. Bladder pressure and EUS-EMG data were recorded through a multi-channel physiological recording device (RM6240; Chengdu Instrument Factory, China) sampled at 8 kHz. After surgery, mice were permitted to recover from anesthesia and resume walking.

For simultaneous recording, all fiber-implanted mice were anesthetized with urethane (1.2 g/kg, i.p.) and continuously infused with room-temperature physiological saline at 30-50 $\mu\text{l/min}$ via the bladder catheter for at least 45 min. Recording or stimulation was performed once regular bladder pressure cycles associated with natural urination events were established. "Filling bladder" was

defined as continuous saline infusion, while “non-filling bladder” was defined as no infusion. Cystometry and EUS-EMG data were captured via commercial acquisition software, alongside monitoring of Ca^{2+} signals and mouse behavior. For cross-correlation analysis, cystometric data, EMG data, and photometry data were first downsampled to 40 Hz and standardized using z-scored. The original EMG data were processed to extract their envelope using MATLAB’s “envelope” function. The original data were merged, and divided into 10 segments, and segments of photometry data were randomly matched with segments of cystometry or envelope EMG data to create shuffled datasets. Cross-correlations of cystometric and photometry data, or EMG and photometry data were calculated using MATLAB’s “xcorr” function, and the peak values of cross-correlations were reported, as shown in Supplementary Figure 3.

For photoactivation experiments performed simultaneously with cystometry and electromyography recording, blue light pulses were delivered periodically (every 30 s) at 25 Hz for 15 ms, lasting 5 s, or randomly at intervals between 20 s and 40 s, with each condition repeated at least 15 times. For photoactivation experiments performed simultaneously with cystometry and urethral electromyography recording under pelvic or pudendal nerve transection conditions, the surgical procedures and pre-recording preparations were the same as those described above, with the following changes: Mice underwent three stages of randomized photostimulation (consisted of 25 Hz, 15 ms, 5-s durations, with intervals between 30 s and 60 s) in sequence under urethane anesthesia (1.2 g/kg, i.p.): an intact nerve period, a pudendal nerve or pelvic nerve transection period, and a period with both pudendal nerve and pelvic nerve transection. All trials were pooled to assess the impact of nerve transection on bladder and sphincter function for each mouse. Cystometric data, including $\Delta P = P_{5 \text{ sec}} - P_{0 \text{ sec}}$ (where $P_{0 \text{ sec}}$ is the pressure at the onset of laser stimulation and $P_{5 \text{ sec}}$ is the pressure at the end of laser stimulation) and the pressure ratio $P_{\text{ratio}} = P_{\text{max (0-5)sec}} / P_{\text{mean (-5-0)sec}}$ (where $P_{\text{max (0-5)sec}}$ is the maximum pressure from laser onset to cessation, and $P_{\text{mean (-5-0)sec}}$ is the average pressure during the 5 seconds preceding laser onset), were calculated using MATLAB. The EUS-EMG data (burst duration and area under the curve) were analyzed using multi-channel physiological recording software. The spectrogram was generated using envelope EMG data around photostimulation in MATLAB, as shown in **Figures 4** and **5**, and **Figure supplement 8**. The onset time of bladder pressure upstroke was identified by finding the maximum of the second derivative of cystometry curves around pressure peaks using a MATLAB script. The onset time of EMG bursting (Kadekawa et al., 2016b; Cheng, 2004) was manually defined as the points at which bursting activity begins. The results are shown in **Figure 8**.

For photoinhibition experiments performed simultaneously with cystometry and electromyography recording, the light was manually triggered at the onset of EMG bursts and delivered in 20 ms pulses at 50 Hz for either 5 s or 60 s. Cystometric data, including $\Delta \text{pressure} = P_{\text{peak}} - P_{\text{min}}$ (where P_{peak} is the peak pressure and P_{min} is the minimum pressure after the peak) and threshold pressure (bladder pressure upstroke), were analyzed manually using the multi-channel physiological recording software. The EUS-EMG data (burst duration, area under the curve, and latency of termination) was analyzed using the multi-channel physiological recording software. The latency of bursting termination is defined as the interval from laser onset to the end of EUS-EMG bursting. The end time of bladder pressure upstroke (termination of the rapid increase in bladder pressure before the end of voiding, denoting bladder relaxation) was identified by finding the minimum of the second derivative of cystometry curves around pressure peaks using a MATLAB script. The end time of EMG bursting was manually defined as the points at which the onset of tonic activity after bursting. The results are shown in **Figures 3** and **8**.

Histology and immunohistochemistry

Following the completion of all experiments, histological verification of fiber implantation or virus injection positions was conducted. Mice were deeply anesthetized with 1% sodium pentobarbital (10 ml/kg), followed by transcardial perfusion with cold 0.9% saline and 4% paraformaldehyde (PFA). The brain or spinal cord (in some experiments) was then extracted and post-fixed overnight

at 4 °C in ice-cold 4% PFA. Coronal brain sections (40 µm) and the thoracolumbar and lumbosacral segments of spinal cord sections (70 µm) were cut using a freezing microtome. For TH immunohistochemistry, free-floating sections were initially incubated in 1% PBST (1% Triton X-100 in PBS, Sigma) for 60 min. The sections were blocked with 10% donkey serum (Sigma) in 0.1% PBST for 2 hours at room temperature. Following blocking, then incubated at 4°C for 24 hours with anti-TH antibodies (1:200 dilution, rabbit, Sigma-Aldrich). After extensive washing with PBS, sections were incubated with a secondary antibody (1:300, Alexa Fluor 488 or 594 donkey anti-rabbit, Invitrogen) at room temperature for 2 hours. Finally, all sections, including the target segments of the spinal cord, were incubated with DAPI (1:1000, Beyotime) for 15 min. Images were acquired using a confocal microscope (TCS SP5, Leica) equipped with × 4, × 10, and × 20 oil objectives, utilizing 405 nm, 488 nm, and 552 nm lasers, or with an Olympus microscope.

Statistical analysis

All data were processed and statistically analyzed using Prism 8 GraphPad, MATLAB, and SPSS 22 software. For unpaired group comparisons, the Wilcoxon rank-sum test was used, and for paired groups, the Wilcoxon signed-rank test was applied, as described in the figure legends. Analysis was performed by investigators blinded to the experiments. The n value reflects the final number of animals in each experiment group, with animals excluded if histological verification of gene expression showed poor or absent. Data are represented as median with 25%-75% percentiles, and statistical significance was defined as * < 0.05, ** < 0.01, *** < 0.001; ns, no significant difference.

Acknowledgements

The authors are grateful to Ms. Jia Lou for her help in composing and editing the layout of the figures. This study was supported by grants from the National Natural Science Foundation of China to X.C. (No. 31925018, 32127801), the National Key R&D Program of China to X.C. (2021YFA0805000), the Suzhou Science and Technology Plan Project (SZS2022008), and the Jiangsu Provincial Big Science Facility Initiative (BM2022010), and the Guangxi Talent Program (“Highland of Innovation Talents”). X.C. is a member of CAS Center for Excellence in Brain Science and Intelligence Technology.

Additional information

Funding

Funder	Grant number	reference	Author
National Key R&D Program of China	2021YFA0805000		Xiaowei Chen
National Natural Science Foundation of China	31925018		Xiaowei Chen
National Natural Science Foundation of China	32127801		Xiaowei Chen
Suzhou Science and Technology Plan Project	SZS2022008		Hongbo Jia
Jiangsu Provincial Big Science Facility Initiative	BM2022010		Hongbo Jia
Guangxi Talent Program	Highland	of	Hongbo Jia
	Innovation Talents		

Author contributions

Project design, J.W.Y. and X.C.; injection and histology, X.L., X.P.L., J.L.(1), L.X.Y., and J.L.(2); behavior experiments, X.L., C.H.Y., and X.W.; nerves transection, X.L.; fiber recording, X.L. C.H.Y., and X.P.L.; electrophysiology, X.L., H.Q., T.L.J., and X.W.; fiber recording, X.L. and L.X.Y.; cystometry and electromyography experiments, X.L., X.P.L., and J.L.(1); data interpretation and analysis, X.L., H.Q., S.S.L., H.B.J., X.Liao, J.W.Y., and X.C.; figure preparation, X.L., H.B.J., J.W.Y. and X.C.; manuscript writing, X.L., H.B.J., J.W.Y. and X.C. with the help of all co-authors. All authors read and commented on the manuscript.

Ethics

All experiment procedures were approved by the Third Military Medical University Animal Care and Use Committee (Approval number: AMUWEC20230061) and were conducted strictly in adherence to established guidelines.

Data availability

Any additional information underlying the findings of this study is available from the corresponding authors upon reasonable request. The supporting data underlying **Figures 1-8** and **Figures supplement 1-9** are provided as Source Data files. Source Data are provided in this paper.

Code availability

The codes supporting the current study have not been deposited in a public repository, but are available from the corresponding author upon request.

Figures supplement

Figure supplement 1.

No change in fluorescence responses of EYFP-labeled $\text{PMC}^{\text{ESR1}+}$ cells during voiding.

(A) Schematics of labeling (left) and representative histology (right) of $\text{PMC}^{\text{ESR1}+}$ cells labeled with EYFP. Scale bar, 100 μm . (B) Cumulative sessions of fluorescence responses aligned to voiding onset ($n = 248$ trials from 9 mice). (C) Average fluorescence traces of $\text{PMC}^{\text{ESR1-EYFP}}$ aligned to voiding onset. The black line and shading represent mean $\pm$ s.e.m., respectively.

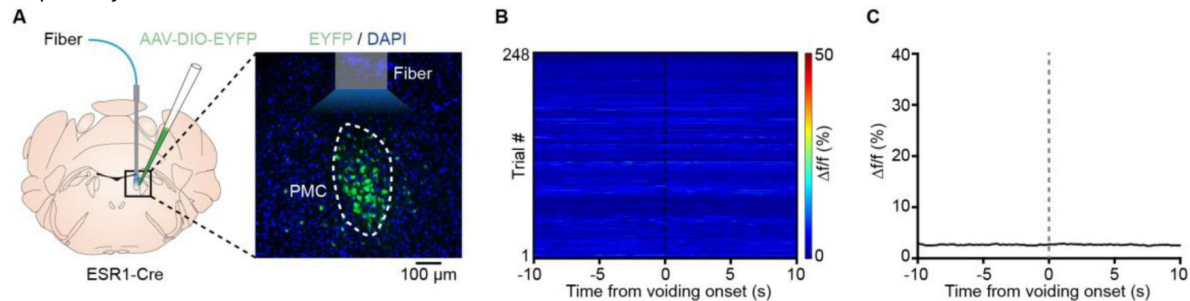
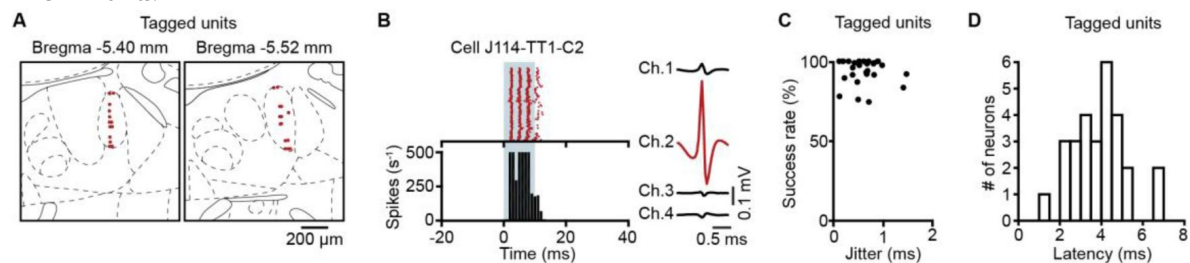


Figure supplement 2.

Identification of recorded $\text{PMC}^{\text{ESR1}+}$ cells in optrode recordings.

(A) Tetraode locations for all recorded $\text{PMC}^{\text{ESR1}+}$ units (red dots, $n = 28$ units from 4 mice). (B) Left: Raster plots (top) and histograms (bottom) showing the firing patterns of representative $\text{PMC}^{\text{ESR1-ChR2}}$ units upon optical stimulation at a power of 10 mW. Right: Waveforms of light-induced spikes from the unit shown on the left. (C) Comparison of success rates and temporal jitter for the first light-induced spike in all recorded $\text{PMC}^{\text{ESR1}+}$ units. (D) Latency distribution for all recorded $\text{PMC}^{\text{ESR1}+}$ units.



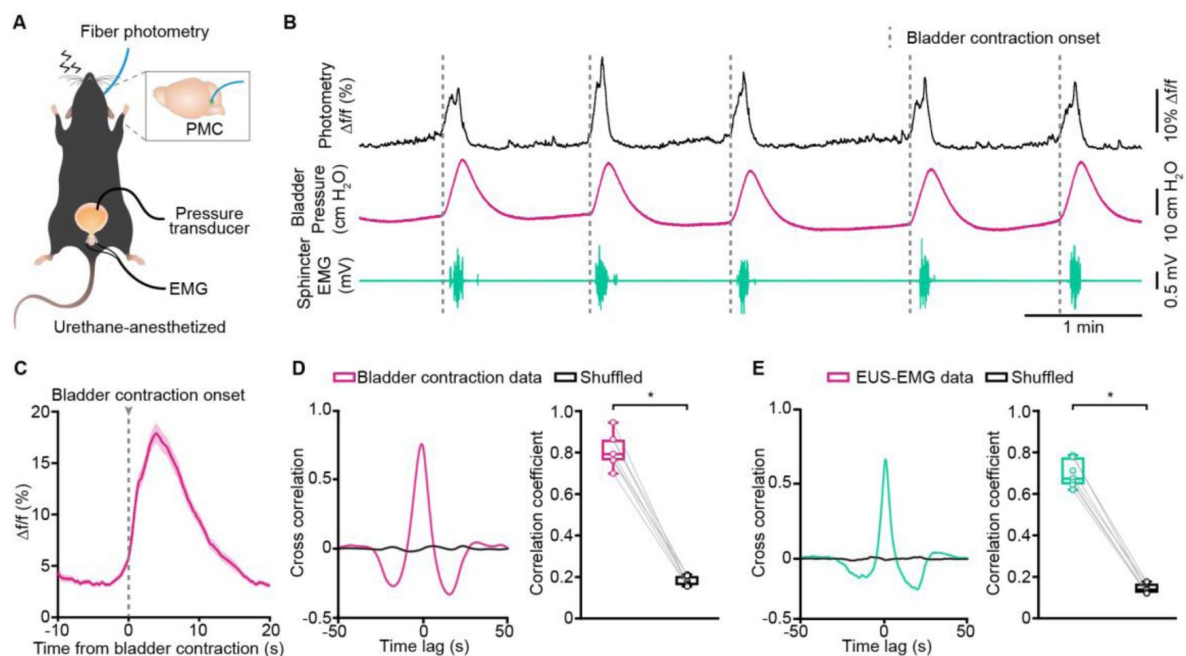


Figure supplement 3.

The activity of PMC^{ESR1+} cells correlates with bladder contraction and sphincter relaxation.

(A) Schematic of fiber photometry recording for PMC^{ESR1+} cells during simultaneous cystometry and urethral electromyography in a urethane-anesthetized mouse. (B) Representative traces showing Ca²⁺ transients (black), bladder pressure (magenta), and EUS-EMG (teal) during fiber photometry recordings, with dashed lines indicating bladder contraction onset. (C) Average Ca²⁺ signals during bladder contraction from all trials (n = 101 trials from 7 mice). The thick line and shading represent mean ± s.e.m., respectively. (D) Cross-correlation (left) and correlation coefficients (right) between Ca²⁺ signals and bladder contraction events compared to shuffled data (n = 7 mice, *P = 0.02, Wilcoxon signed-rank test). (E) Cross-correlation (left) and correlation coefficients (right) between Ca²⁺ signals and EUS-EMG bursting events compared to shuffled data (n = 7 mice, *P = 0.02, Wilcoxon signed-rank test). For all data points in (D, E), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

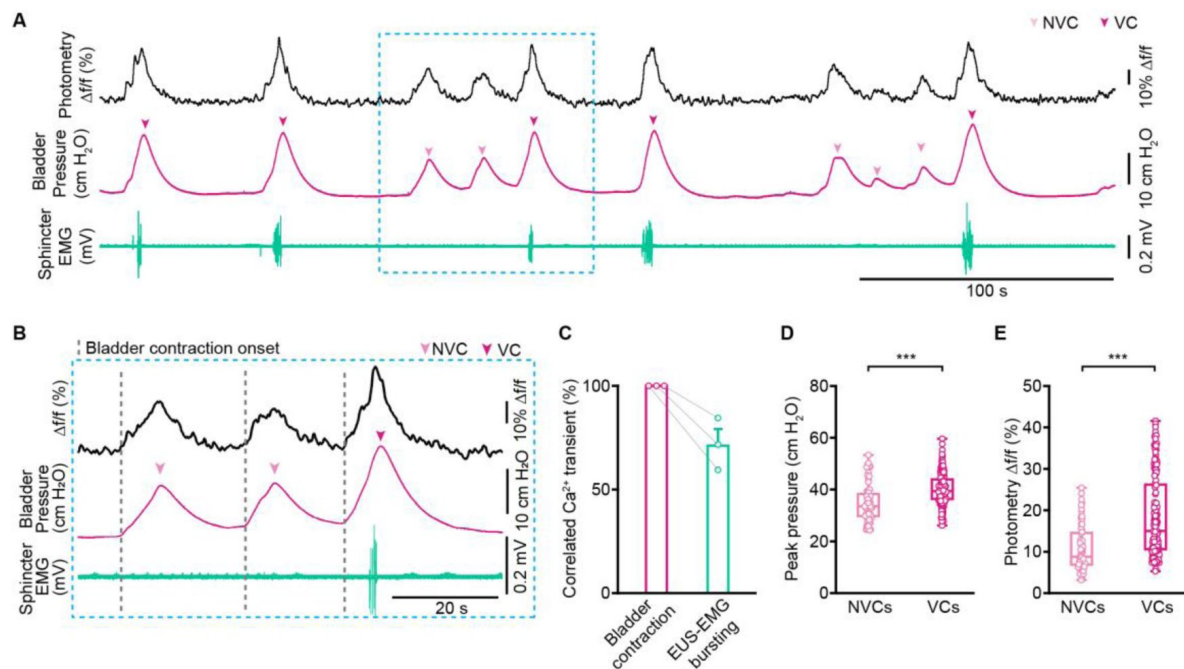


Figure supplement 4.

No voiding contractions (NVCs) correlate with Ca^{2+} signals of $\text{PMC}^{\text{ESR1}+}$ cells.

(A) Representative traces of Ca^{2+} transients (black), bladder pressure (magenta), and EUS-EMG (teal) for no voiding contractions (NVCs) and voiding contractions (VCs). (B) Expanded portions of no voiding contractions (NVCs, light magenta arrow) and voiding contractions (VC, dark magenta arrow) from (A) (blue dashed box). (C) Correlation rate of Ca^{2+} transient with bladder contraction and EUS-EMG bursting events ($n = 3$ mice per group). (D, E) Comparison of peak bladder pressure (D) and the amplitude of Ca^{2+} transients (E) between NVCs and VCs (NVCs: $n = 62$ events from 3 mice, VCs: $n = 167$ events from 3 mice; *** $P = 6.03 \times 10^{-7}$, *** $P = 6.8 \times 10^{-9}$, respectively, Wilcoxon rank-sum test). For all data points in (D, E), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

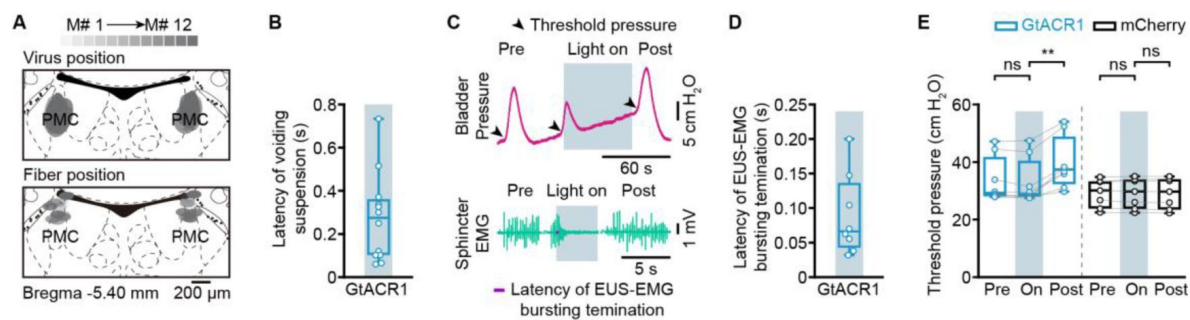


Figure supplement 5.

Acute photoinhibition (60 s) of PMC^{ESR1+} cells operates the bladder and sphincter to suspend voiding, related to Figure 2.

(A) Overlay of viral expression areas (top) and fiber positions (bottom) from ESR1-Cre mice labeled with GtACR1 (n = 12 mice). (B) Latency of voiding suspension after light activation. (C) Representative raw traces of bladder pressure and EUS-EMG. (D) Latency of sphincter bursting termination after light activation (n = 8 mice). (E) Bladder threshold pressure during voiding before ('Pre'), during ('On'), and after ('Post') photoinhibition in PMC^{ESR1}-GtACR1 (n = 8 mice) and PMC^{ESR1}-mCherry (n = 7 mice) groups (from left to right: $P = 0.06$, $**P = 7.8e-3$, $P = 0.9$, $P = 0.4$, respectively; n.s., not significant; Wilcoxon signed-rank test). For all data points in (B, D) and (E), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

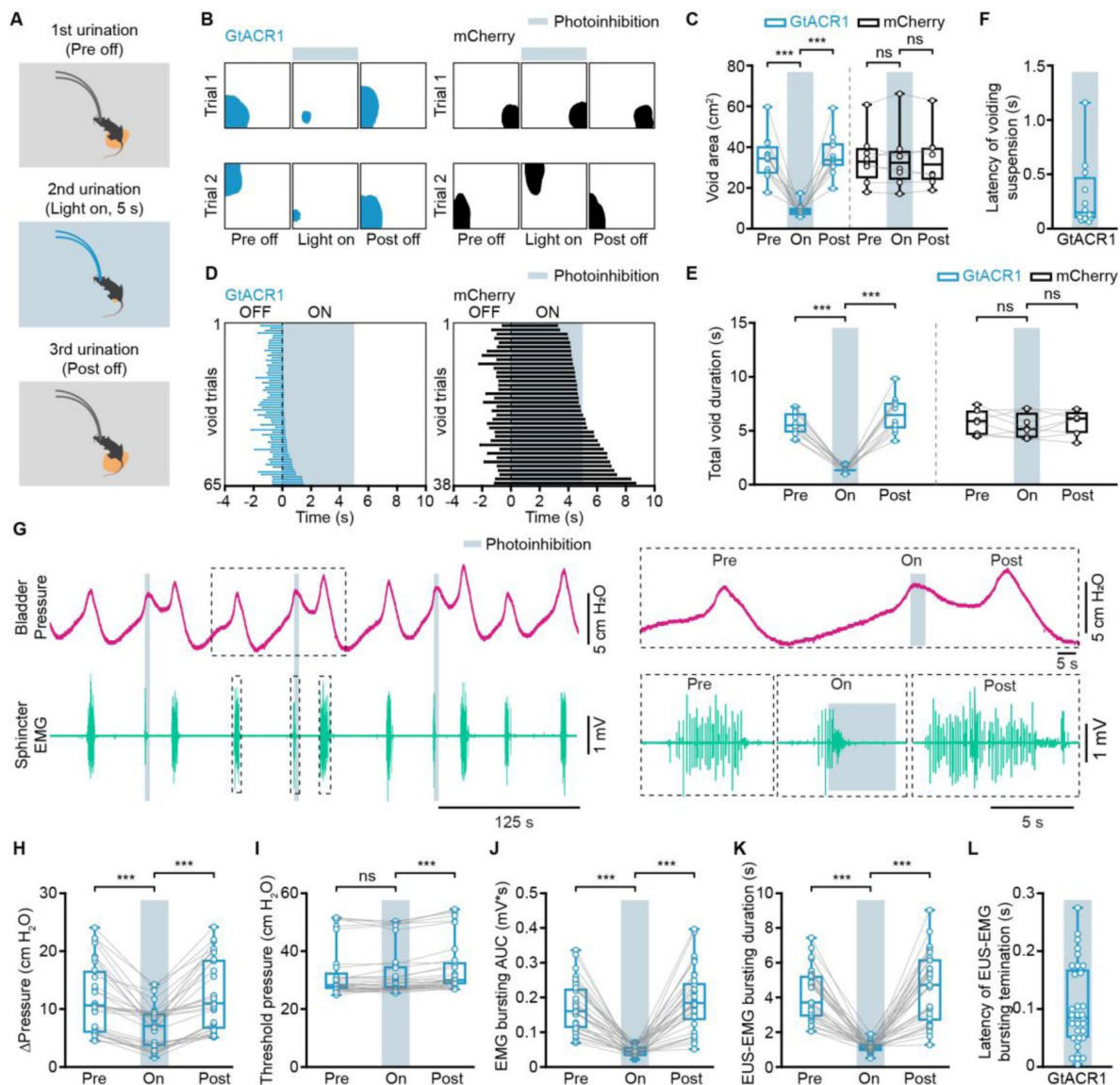


Figure supplement 6.

5-s photoinhibition of PMC^{ESR1+} cells suppresses bladder contraction and sphincter relaxation to suspend ongoing voiding.

(A) Experimental design for 5 s photoinhibition of PMC^{ESR1+} cells in a freely moving mouse. (B, C) Representative images (B, blue and black shading) and quantification (C) of the void area before, during, and after 5 s photoinhibition in PMC^{ESR1-GtACR1} (n = 12 mice) and PMC^{ESR1-mCherry} (n = 8 mice) groups (from left to right: ***P = 4.9e-4, ***P = 4.9e-4, P = 0.7, P = 0.9, respectively; n.s., not significant; Wilcoxon signed-rank test). (D) Cumulative trials of voiding duration during 5 s photoinhibition in PMC^{ESR1-GtACR1} (blue bar, n = 65 trials from 12 mice) and PMC^{ESR1-mCherry} groups (black bar, n = 38 trials from 8 mice), ordered by increasing voiding epoch time with the laser on. (E) Voiding duration before, during, and after 5 s photoinhibition in PMC^{ESR1-GtACR1} (n = 12 mice) and PMC^{ESR1-mCherry} (n = 8 mice) groups (from left to right: ***P = 4.9e-4, ***P = 4.9e-4, P = 0.3, P = 0.4, respectively; n.s., not significant; Wilcoxon signed-rank test). (F) Latency of urination suspension after light activation. (G) Representative traces (left) and expanded portions (right, from the dashed box in the left panel) of bladder pressure (magenta) and EUS-EMG (teal) before, during, and after 5 s photoinhibition in PMC^{ESR1-GtACR1} individual. (H-L) Quantification of the effect of 5 s photoinhibition (n = 33 trials from 6 mice) on bladder pressure and EUS-EMG: Δpressure (H, ***P = 5.4e-7), threshold pressure (I, P = 0.98, ***P = 1.9e-6, respectively), EUS-EMG bursting AUC (J, ***P = 5.4e-7), and EUS-EMG bursting duration (K, ***P = 5.4e-7; n.s., not significant; Wilcoxon signed-rank test). (L) Latency of sphincter bursting termination after 5 s photoinhibition. For all data points in (C), (E), (F), and (H-L), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

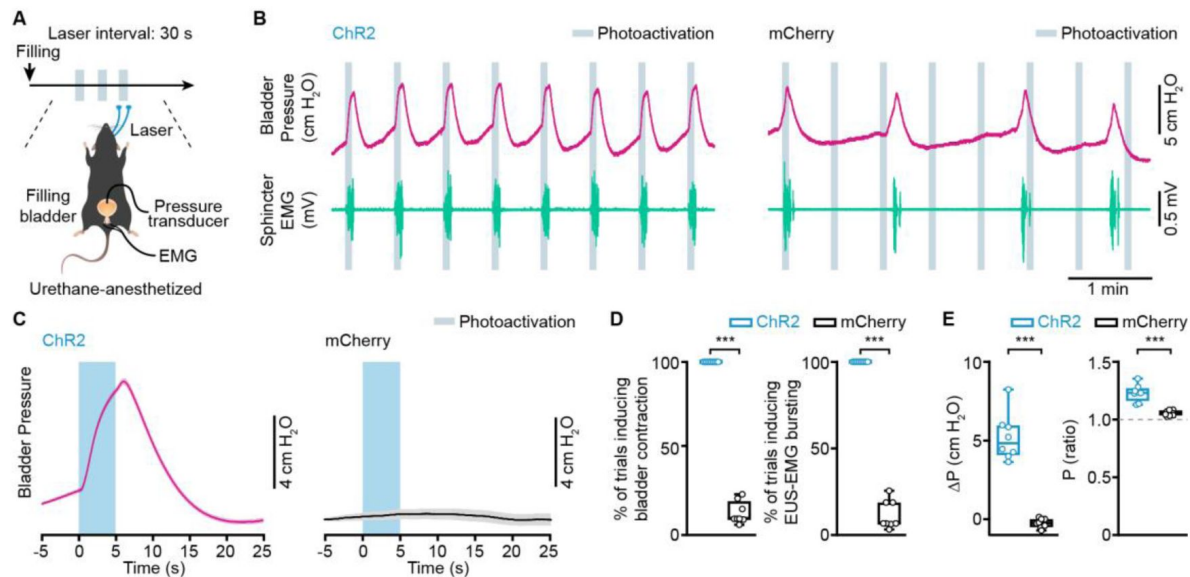


Figure supplement 7.

Regular interval photoactivation of PMC^{ESR1+} cells induces both bladder contraction and external urethral sphincter relaxation.

(A) Timeline (top) and schematic (bottom) for regular interval photoactivation of PMC^{ESR1+} cells during simultaneous cystometry and urethral electromyography recording with a filled bladder. (B) Representative raw traces of bladder pressure (magenta) and EUS-EMG (teal) around the photoactivation timepoint in PMC^{ESR1-ChR2} (left) and PMC^{ESR1-mCherry} (right) individuals. (C) Average bladder pressure around photoactivation in PMC^{ESR1-ChR2} (left, n = 8 mice) and PMC^{ESR1-mCherry} (right, n = 8 mice) groups. The thick line and shading represent mean ± s.e.m., respectively. (D, E) Quantification of the photoactivation effect on the bladder detrusor and urethra sphincter in PMC^{ESR1-ChR2} (n = 8 mice) or PMC^{ESR1-mCherry} (n = 8 mice) groups: the percentage of bladder contraction (D, left), the percentage of EUS-EMG bursting (D, right), ΔP (cm H₂O) (E, left), and bladder pressure ratio (E, right; ***P = 1.6e-4 for D and E, Wilcoxon rank-sum test). For all data points in (D, E), whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

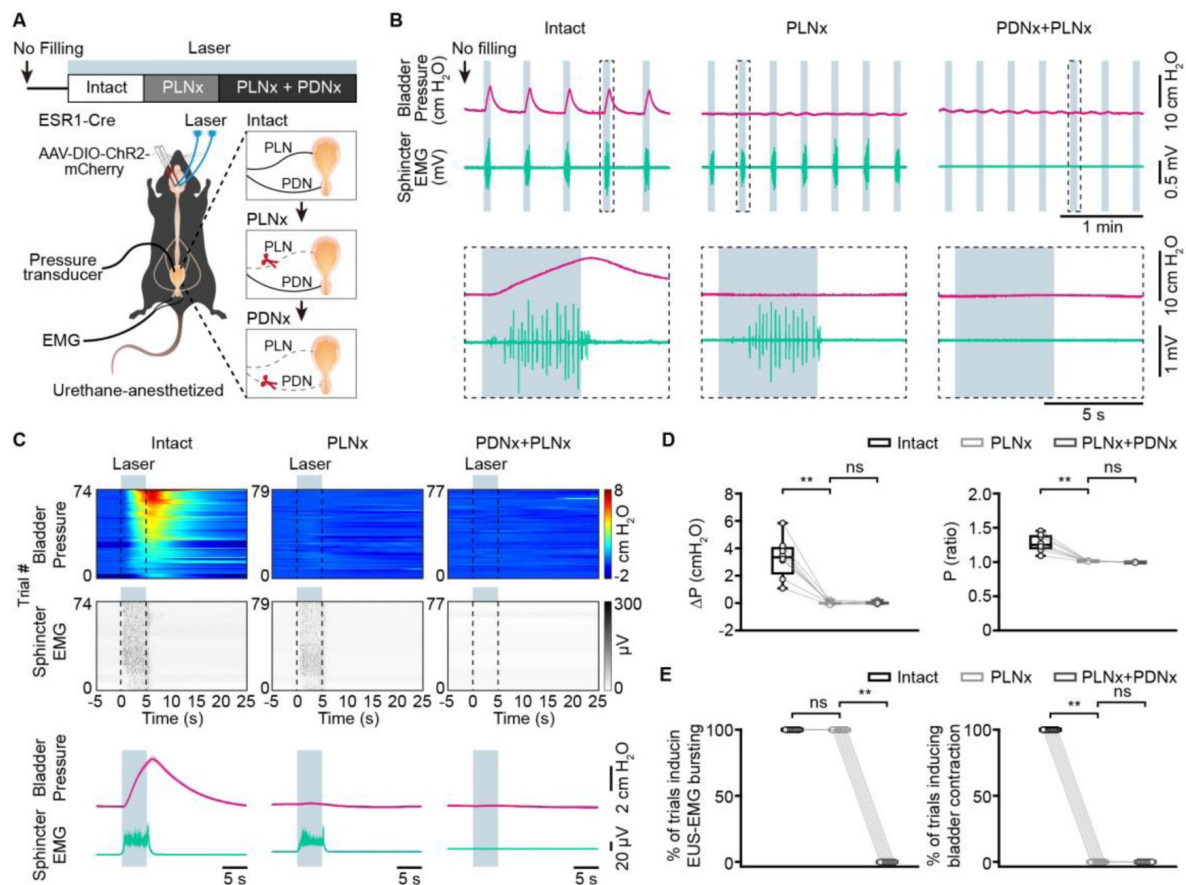


Figure supplement

8. Transection of the pelvic nerves does not alter bladder pressure in an unfilled bladder during PMC^{ESR1-ChR2} photoactivation.

(A) Timeline (top) and schematic (bottom) for PMC^{ESR1-ChR2} photoactivation during simultaneous cystometry and urethral electromyography recordings in a non-filled bladder, with PLNx performed first. PLNx: pelvic nerve transection; PDNx: pudendal nerve transection. (B) Representative traces (top) and expanded portions (bottom, from the dashed box in the top panel) showing bladder pressure (magenta) and EUS-EMG (teal) during PMC^{ESR1-ChR2} photoactivation in an unfilled bladder with PLNx performed first. (C) Heatmap (top) and average traces (bottom, thick line, and shading represent mean $\pm$ s.e.m., respectively) of sorted bladder pressure and EUS-EMG around photoactivation timepoint for all unfilled bladder trials with PLNx performed first ($n = 8$ mice per group). (D, E) Quantification of bladder pressure change (ΔP , D, left), bladder pressure ratio (D, right), the percentage of photoactivation-associated bladder contraction (E, left), and the percentage of photoactivation-associated EUS-EMG bursting (E, right) upon photoactivation for the PLNx-first experiment from C ($n = 8$ mice per group, from left to right, D: $**P = 7.8e-3$, $P = 0.8$, $**P = 7.8e-3$, $P = 0.1$, respectively; E: $P = 1$, $**P = 7.8e-3$, $**P = 7.8e-3$, $P = 1$, respectively; n.s., not significant; Wilcoxon signed-rank test). For all data points in (D, E) whisker-box plots indicate the median with the 25%-75% percentile as the box, and whiskers represent the minimum and maximum values.

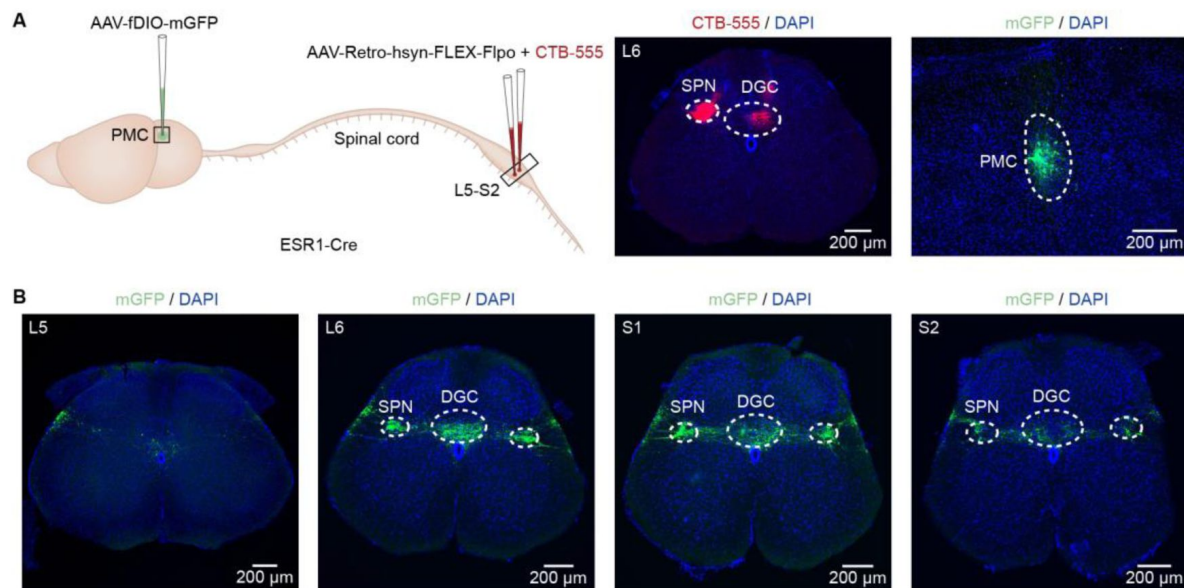


Figure supplement 9.

PMC^{ESR1+} cells projection to the SPN and DGC in the spinal cord.

(**A**) Left: Schematic of labeling. Middle and right: Representative histological images showing CTB-555 expression in the lumbar spinal cord (middle) and mGFP expression in the PMC (right). Scale bars: 200 μ m. (**B**) Axonal projections of PMC^{ESR1}-mGFP cells in the lumbar spinal cord from L5 to S2 levels (n = 3 mice). Scale bars: 200 μ m. Abbreviations: SPN, sacral parasympathetic nucleus; DGC, dorsal gray commissure.

References

- Abud EM, Ichiyama RM, Havton LA, Chang HH (2015) **Spinal stimulation of the upper lumbar spinal cord modulates urethral sphincter activity in rats after spinal cord injury** *Am J Physiol Renal Physiol* **308**:F1032–40 <https://doi.org/10.1152/ajprenal.00573.2014>
- Andersson K.-E (2004) **Urinary Bladder Contraction and Relaxation: Physiology and Pathophysiology** *Physiol Reviews* **84**:935–986 <https://doi.org/10.1152/physrev.00038.2003>
- Aoki Y, Brown HW, Brubaker L, Cornu JN, Daly JO, Cartwright R (2017) **Urinary incontinence in women** *Nature Reviews Disease Primers* **3** <https://doi.org/10.1038/nrdp.2017.42>
- Baram TZ, Ressler K, Gunn BG, Molet J, Chen Y (2015) **Diversity of Reporter Expression Patterns in Transgenic Mouse Lines Targeting Corticotropin-Releasing Hormone-Expressing Neurons** *Endocrinology* **156**:4769–4780 <https://doi.org/10.1210/en.2015-1673>
- Benarroch EE (2010) **Neural control of the bladder Recent advances and neurologic implications** *Neurology* **75**:1839–1846 <https://doi.org/10.1212/WNL.0b013e3181fdabba>
- Biallostowski BT, van Koevinge GA, van Kerrebroeck PE, Gillespie JI, de Wachter SG (2011) **Nonvoiding Activity of the Guinea Pig Bladder** *Journal of Urology* **186**:721–727 <https://doi.org/10.1016/j.juro.2011.03.123>
- Chang H, Yeh J, Ichiyama R, Rodriguez L, Havton L (2018) **Mapping and neuromodulation of lower urinary tract function using spinal cord stimulation in female rats** *Exp Neurol* **305**:26–32 <https://doi.org/10.1016/j.expneurol.2018.03.007>
- Chang HY, Cheng CL, Chen JJJ, de Groat WC (2007) **Serotonergic drugs and spinal cord transections indicate that different spinal circuits are involved in external urethral sphincter activity in rats** *American Journal of Physiology-Renal Physiology* **292**:F1044–F1053 <https://doi.org/10.1152/ajprenal.00175.2006>
- Chen KS *et al.* (2019) **Targeted intraspinal injections to assess therapies in rodent models of neurological disorders** *Nat Protoc* **14**:331–349 <https://doi.org/10.1038/s41596-018-0095-5>
- Cheng C (2004) **The role of capsaicin-sensitive afferent fibers in the lower urinary tract dysfunction induced by chronic spinal cord injury in rats** *Experimental Neurology* **187**:445–454 <https://doi.org/10.1016/j.expneurol.2004.02.014>
- Cho H, Kang T, Chang J, Choi Y, Park M, Choi K, Sung S, Park K, Jung D (2015) **Neuroanatomical correlation of urinary retention in lateral medullary infarction** *Annals of Neurology* **77**:726–733 <https://doi.org/10.1002/ana.24379>
- de Groat WC (2009) **Integrative control of the lower urinary tract: preclinical perspective** *British Journal of Pharmacology* **147** <https://doi.org/10.1038/sj.bjp.0706604>
- de Groat WC, Griffiths D, Yoshimura N. (2015) **Neural Control of the Lower Urinary Tract** *Compr Physiol* **5**:327–396 <https://doi.org/10.1002/cphy.c130056>

- Desjardins C, Maruniak J, Bronson F (1973) **Social Rank in House Mice: Differentiation Revealed by Ultraviolet Visualization of Urinary Marking Patterns** *Science* **182**:939–941 <https://doi.org/10.1126/science.182.4115.939>
- Drake MJ, Williams J, Bijos DA (2014) **Voiding dysfunction due to detrusor underactivity: an overview** *Nature Reviews Urology* **11**:454–464 <https://doi.org/10.1038/nrurol.2014.156>
- Fang YY, Yamaguchi T, Song SC, Tritsch NX, Lin D (2018) **A Hypothalamic Midbrain Pathway Essential for Driving Maternal Behaviors** *Neuron* **98**:192–207 <https://doi.org/10.1016/j.neuron.2018.02.019>
- Fowler C, Griffiths D, de Groat W. (2008) **The neural control of micturition** *Nat Rev Neurosci* **9**:453–466 <https://doi.org/10.1038/nrn2401>
- Griffiths D (2015) **Neural control of micturition in humans: a working model** *Nature Reviews Urology* **12**:695–705 <https://doi.org/10.1038/nrurol.2015.266>
- Griffiths D, Derbyshire S, Stenger A N. R (2005) **Brain control of normal and overactive bladder** *J Urol* **174**:1862–1867 <https://doi.org/10.1097/01.ju.0000177450.34451.97>
- Hou XH *et al.* (2016) **Central Control Circuit for Context-Dependent Micturition** *Cell* **167**:73–86 <https://doi.org/10.1016/j.cell.2016.08.073>
- Hyun M, Taranda J, Radeljic G, Miner L, Wang W, Ochandarena N, Huang KW, Osten P, Sabatini B (2021) **Social isolation uncovers a brain-wide circuit underlying context-dependent territory-covering micturition behavior** *Proc Natl Acad Sci U S A* **118** <https://doi.org/10.1101/798132>
- Ito H, Sales AC, Fry CH, Kanai AJ, Drake MJ, Pickering AE (2020) **Probabilistic, spinally-gated control of bladder pressure and autonomous micturition by Barrington’s nucleus CRH neurons** *eLife* **9** <https://doi.org/10.7554/eLife.56605>
- Jin WK, Su JK, Jong MP, Yong GN, Khae HK (2020) **Past Present, and Future in the Study of Neural Control of the Lower Urinary Tract** **24**:191–199 <https://doi.org/10.5213/inj.2040318.159>
- Kadekawa K, Yoshimura N, Majima T, Wada N, Shimizu T, Birder LA, Kanai AJ, de Groat WC, Sugaya K, Yoshiyama M. (2016) **Characterization of bladder and external urethral activity in mice with or without spinal cord injury--a comparison study with rats** *Am J Physiol Regul Integr Comp Physiol* **310**:R752–8 <https://doi.org/10.1152/ajpregu.00450.2015>
- Kadekawa K, Yoshimura N, Majima T, Wada N, Shimizu T, Birder LA, Kanai AJ, de Groat WC, Sugaya K, Yoshiyama M. (2016) **Characterization of bladder and external urethral activity in mice with or without spinal cord injury—a comparison study with rats** *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* **310**:R752–R758 <https://doi.org/10.1152/ajpregu.00450.2015>
- Karigo T, Kennedy A, Yang B, Liu M, Tai D, Wahle I, Anderson D (2021) **Distinct hypothalamic control of same- and opposite-sex mounting behaviour in mice** *Nature* **589**:258–263 <https://doi.org/10.1038/s41586-020-2995-0>
- Kaur Angeldeep W *et al.* (2014) **Murine Pheromone Proteins Constitute a Context-Dependent Combinatorial Code Governing Multiple Social Behaviors** *Cell* **157**:676–688 <https://doi.org/10.1016/j.cell.2014.02.025>

- Kawatani M, de Groat WC, Itoi K, Uchida K, Sakimura K, Yamanaka A, Yamashita T, Kawatani M. (2021) **Downstream projection of Barrington's nucleus to the spinal cord in mice** *Journal of Neurophysiology* **126**:1959–1977 <https://doi.org/10.1152/jn.00026.2021>
- Keller JA, Chen J, Simpson S, Wang EH-J, Lilascharoen V, George O, Lim BK, Stowers L (2018) **Voluntary urination control by brainstem neurons that relax the urethral sphincter** *Nature Neuroscience* **21**:1229–1238 <https://doi.org/10.1038/s41593-018-0204-3>
- Khorramirouz R, Mozafarpour S, Kameli SM, Ladi Seyedian SS, Oveisi N, Rahimi Z, Alijani M, Kajbafzadeh AM (2016) **A Novel Method of Urinary Sphincter Deficiency: Serial Histopathology Evaluation in a Rat Model of Urinary Incontinence** *Anat Rec (Hoboken)* **299**:173–80 <https://doi.org/10.1002/ar.23291>
- Langdale CL, Grill WM (2016) **Phasic activation of the external urethral sphincter increases voiding efficiency in the rat and the cat** *Experimental Neurology* **285**:173–181 <https://doi.org/10.1016/j.expneurol.2016.05.030>
- Lee C, Lee J, Park J, Na H, Shin J, Na Y, Kim K (2021) **Sophisticated regulation of micturition: review of basic neurourology** *J Exerc Rehabil* **17**:295–307 <https://doi.org/10.12965/jer.2142594.297>
- Lee H, Kim D, Remedios R, Anthony T, Chang A, Madisen L, Zeng H, Anderson D (2014) **Scalable control of mounting and attack by Esr1+ neurons in the ventromedial hypothalamus** *Nature* **509**:627–632 <https://doi.org/10.1038/nature13169>
- Liu M, Kim D, Zeng H, Anderson D (2022) **Make war not love: The neural substrate underlying a state-dependent switch in female social behavior** *Neuron* **110**:841–856 <https://doi.org/10.1016/j.neuron.2021.12.002>
- Mallory B (1991) **Pharmacological modulation of the pontine micturition center** *Brain Research* **546**:310–320 [https://doi.org/10.1016/0006-8993\(91\)91495-m](https://doi.org/10.1016/0006-8993(91)91495-m)
- Malykhina AP (2017) **How the brain controls urination** *eLife* **6** <https://doi.org/10.7554/eLife.33219>
- Manohar A, Curtis AL, Zderic SA, Valentino RJ (2017) **Brainstem network dynamics underlying the encoding of bladder information** *Elife* **6** <https://doi.org/10.7554/eLife.29917>
- Milsom I, Coyne KS, Nicholson S, Kvasz M, Chen C-I, Wein AJ (2014) **Global Prevalence and Economic Burden of Urgency Urinary Incontinence: A Systematic Review** *European Urology* **65**:79–95 <https://doi.org/10.1016/j.eururo.2013.08.031>
- Morrison JFB (2008) **The discovery of the pontine micturition centre by F.J. F. Barrington.** *Experimental Physiology* **93**:742–745 <https://doi.org/10.1113/expphysiol.2007.038976>
- Mukhopadhyay S, Stowers L (2020) **Choosing to urinate. Circuits and mechanisms underlying voluntary urination** *Current Opinion in Neurobiology* **60**:129–135 <https://doi.org/10.1016/j.conb.2019.11.004>
- Noto H, Roppolo J, Steers W, de Groat W. (1989) **Excitatory and inhibitory influences on bladder activity elicited by electrical stimulation in the pontine micturition center in the rat** *Brain Research* **492**:99–115 [https://doi.org/10.1016/0006-8993\(89\)90893-7](https://doi.org/10.1016/0006-8993(89)90893-7)

Peh WYX, Mogan R, Thow XY, Chua SM, Rusly A, Thakor NV, Yen SC (2018) **Novel Neurostimulation of Autonomic Pelvic Nerves Overcomes Bladder-Sphincter Dyssynergia** *Front Neurosci* **12** <https://doi.org/10.3389/fnins.2018.00186>

Qin H *et al.* (2018) **A Visual-Cue-Dependent Memory Circuit for Place Navigation** *Neuron* **99**:47–55 <https://doi.org/10.1016/j.neuron.2018.05.021>

Qin H *et al.* (2022) **REM sleep-active hypothalamic neurons may contribute to hippocampal social-memory consolidation** *Neuron* **110**:4000–4014 <https://doi.org/10.1016/j.neuron.2022.09.004>

Rana S, Alom F, Martinez RC, Fuller DD, Mickle AD (2024) **Acute ampakines increase voiding function and coordination in a rat model of SCI** *Elife* **12** <https://doi.org/10.7554/eLife.89767>

Rao Y *et al.* (2022) **Ventrolateral Periaqueductal Gray Neurons Are Active During Urination** *Frontiers in Cellular Neuroscience* **16** <https://doi.org/10.3389/fncel.2022.865186>

Sakakibara Ryuji (2015) **Lower urinary tract dysfunction in patients with brain lesions** *Neurology of Sexual and Bladder Disorders* **130**:269–287 <https://doi.org/10.1016/b978-0-444-63247-0.00015-8>

Sartori AM, Hofer AS, Scheuber MI, Rust R, Kessler TM, Schwab ME (2022) **Slow development of bladder malfunction parallels spinal cord fiber sprouting and interneurons' loss after spinal cord transection** *Exp Neurol* **348** <https://doi.org/10.1016/j.expneurol.2021.113937>

Seseke S, Leitsmann C, Hijazi S, Trojan L, Dechent P (2019) **Functional MRI in patients with detrusor sphincter dyssynergia: Is the neural circuit affected?** *Neurourology and Urodynamics* **38**:2104–2111 <https://doi.org/10.1002/nau.24112>

Stoffel J (2016) **Detrusor sphincter dyssynergia: a review of physiology, diagnosis, and treatment strategies** *Transl Androl Urol* **5**:127–135 <https://doi.org/10.3978/j.issn.2223-4683.2016.01.08>

Sugaya K, De Groat W. (1994) **Micturition reflexes in the in vitro neonatal rat brain stem-spinal cord-bladder preparation** *Am J Physiol* **266**:R658–67 <https://doi.org/10.1152/ajpregu.1994.266.3.r658>

Taweel AI, Waleed Seyam, Raouf (2015) **Neurogenic bladder in spinal cord injury patients** *Research and Reports in Urology* <https://doi.org/10.2147/rru.S29644>

Valentino RJ, Wood SK, Wein AJ, Zderic SA (2010) **The bladder–brain connection: putative role of corticotropin-releasing factor** *Nature Reviews Urology* **8**:19–28 <https://doi.org/10.1038/nrurol.2010.203>

Van Batavia JP, Butler S, Lewis E, Fesi J, Canning DA, Vicini S, Valentino RJ, Zderic SA. (2021) **Corticotropin-Releasing Hormone from the Pontine Micturition Center Plays an Inhibitory Role in Micturition** *The Journal of Neuroscience* **41**:7314–7325 <https://doi.org/10.1523/jneurosci.0684-21.2021>

Vanderhorst VGJM, Gustafsson JÅ, Ulfhake B (2005) **Estrogen receptor- α and - β immunoreactive neurons in the brainstem and spinal cord of male and female mice: Relationships to monoaminergic, cholinergic, and spinal projection systems** *Journal of Comparative Neurology* **488**:152–179 <https://doi.org/10.1002/cne.20569>

- Verstegen A, Vanderhorst V, Gray P, Zeidel M, Geerling J (2017) **Barrington's nucleus: Neuroanatomic landscape of the mouse "pontine micturition center"** *J Comp Neurol* **525**:2287–2309 <https://doi.org/10.1002/cne.24215>
- Verstegen AMJ *et al.* (2019) **Non-Crh Glutamatergic Neurons in Barrington's Nucleus Control Micturition via Glutamatergic Afferents from the Midbrain and Hypothalamus** *Current Biology* **29**:2775–2789 <https://doi.org/10.1016/j.cub.2019.07.009>
- Vincent S, Satoh K (1984) **Corticotropin-releasing factor (CRF) immunoreactivity in the dorsolateral pontine tegmentum: further studies on the micturition reflex system** *Brain Res* **308**:387–391 [https://doi.org/10.1016/0006-8993\(84\)91085-0](https://doi.org/10.1016/0006-8993(84)91085-0)
- Wood SK, Baez MA, Bhatnagar S, Valentino RJ (2009) **Social stress-induced bladder dysfunction: potential role of corticotropin-releasing factor** *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* **296**:R1671–R1678 <https://doi.org/10.1152/ajpregu.91013.2008>
- Xiao W *et al.* (2021) **Neural circuit control of innate behaviors** *Science China Life Sciences* **65**:466–499 <https://doi.org/10.1007/s11427-021-2043-2>
- Yang X, Chen Q, Jian T, Du H, Jin W, Liang M, Wang R, Chen X, Liao X, Qin H. (2023) **Optrode recording of an entorhinal-cortical circuit in freely moving mice** *Biomed Opt Express* **14**:1911–1922 <https://doi.org/10.1364/BOE.487191>
- Yao J *et al.* (2018) **A corticopontine circuit for initiation of urination** *Nature Neuroscience* **21**:1541–1550 <https://doi.org/10.1038/s41593-018-0256-4>
- Zare A, Jahanshahi A, Rahnama'i M, Schipper S, van Koeveeringe G. (2019) **The Role of the Periaqueductal Gray Matter in Lower Urinary Tract Function** *Mol Neurobiol* **56**:920–934 <https://doi.org/10.1007/s12035-018-1131-8>
- Zderic SA (2019) **Neuroscience: A New Golden Age for Neurourology** *Current Biology* **29**:R880–R883 <https://doi.org/10.1016/j.cub.2019.08.009>

Author information

Xing Li

Advanced Institute for Brain and Intelligence, School of Physical Science and Technology, Guangxi University, Nanning, China

Xianping Li

Brain Research Center and State Key Laboratory of Trauma and Chemical Poisoning, Third Military Medical University, Chongqing, China, Department of Urology, PLA Naval Medical Center, Naval Medical University, Shanghai, China

Jun Li

Advanced Institute for Brain and Intelligence, School of Physical Science and Technology, Guangxi University, Nanning, China

Han Qin

LFC Laboratory and Chongqing Institute for Brain and Intelligence, Guangyang Bay Laboratory, Chongqing, China

Shanshan Liang

Brain Research Center and State Key Laboratory of Trauma and Chemical Poisoning, Third Military Medical University, Chongqing, China

Jun Li

Brain Research Center and State Key Laboratory of Trauma and Chemical Poisoning, Third Military Medical University, Chongqing, China

Tingliang Jian

Brain Research Center and State Key Laboratory of Trauma and Chemical Poisoning, Third Military Medical University, Chongqing, China

Xia Wang

Center for Neurointelligence, School of Medicine, Chongqing University, Chongqing, China

Lingxuan Yin

Advanced Institute for Brain and Intelligence, School of Physical Science and Technology, Guangxi University, Nanning, China

Chunhui Yuan

Department of Urology, PLA Naval Medical Center, Naval Medical University, Shanghai, China

Xiang Liao

Center for Neurointelligence, School of Medicine, Chongqing University, Chongqing, China

Hongbo Jia

Advanced Institute for Brain and Intelligence, School of Physical Science and Technology, Guangxi University, Nanning, China, Leibniz Institute for Neurobiology, Magdeburg, Germany, Suzhou Institute of Biomedical Engineering and Technology, Chinese Academy of Sciences, Suzhou, China

ORCID iD: [0000-0003-1585-2161](https://orcid.org/0000-0003-1585-2161)

For correspondence: jiahb@sibet.ac.cn

Xiaowei Chen

Brain Research Center and State Key Laboratory of Trauma and Chemical Poisoning, Third Military Medical University, Chongqing, China, LFC Laboratory and Chongqing Institute for Brain and Intelligence, Guangyang Bay Laboratory, Chongqing, China

ORCID iD: [0000-0003-0906-6666](https://orcid.org/0000-0003-0906-6666)

For correspondence: xiaowei_chen@tmmu.edu.cn

Jiwei Yao

Department of Urology, PLA Naval Medical Center, Naval Medical University, Shanghai, China, Center for Neurointelligence, School of Medicine, Chongqing University, Chongqing, China

ORCID iD: [0009-0003-5063-6584](https://orcid.org/0009-0003-5063-6584)

For correspondence: jiwei_yao89@cqu.edu.cn

Editors

Reviewing Editor

Alexander Chesler

National Institutes of Health, Bethesda, United States of America

Senior Editor

Timothy Behrens

University of Oxford, Oxford, United Kingdom

Reviewer #1 (Public review):

Summary:

Urination requires precise coordination between the bladder and external urethral sphincter (EUS), while the neural substrates controlling this coordination remain poorly understood. In this study, Li et al. identify estrogen receptor 1-expressing neurons (ESR1+) in Barrington's nucleus as key regulators that faithfully initiate or suspend urination. Results from peripheral nerve lesions suggest that BarEsr1 neurons play independent roles in controlling bladder contraction and relaxation of the EUS. Finally, the authors performed region-specific retrograde tracing, claiming that distinct populations of BarEsr1 neurons target specific spinal nuclei involved in regulating the bladder and EUS, respectively.

Strength:

Overall, the work is of high quality. The authors integrate several cutting-edge technologies and sophisticated, thorough analyses, including opto-tagged single unit recordings, combined optogenetics, and urodynamics, particularly those following distinct peripheral nerve lesions.

Weakness:

(1) My major concern is the novelty of this study. Keller et al. 2018 have shown that BarEsr1 neurons are active during urination and play an essential role in relaxing the external urethral sphincter (EUS). Minimally, substantial content that merely confirms previous findings (e.g. Figures 1A-E; Figures 3A-E) should be moved to the supplementary datasets.

(2) I also have concerns regarding the results showing that the inactivation of BarEsr1 neurons led to the cessation of EUS muscle firing (Figures 2G and S5C). As shown in the cartoon illustration of Figure 8, spinal projections of BarEsr1 neurons contact interneurons (presumably inhibitory) that innervate motor neurons, which in turn excite the EUS. I would therefore expect that the inactivation of BarEsr1 should shift the EUS firing pattern from phasic (as relaxation) to tonic (removal of relaxation), rather than stopping their firing entirely. Could the authors comment on this and provide potential reasons or mechanisms for this finding?

(3) Current evidence is insufficient to support the claim that the majority of BarEsr1 neurons innervate the SPN but not DGC. The current spinal images are uninformative, as the fluorescence reflects the distribution of Esr1- or Crh-expressing neurons in the spinal cord, along with descending BarEsr1 or BarCrh axons. Given the close anatomical proximity of these two nuclei, a more thorough histological analysis is required to demonstrate that the spinal injections were accurately confined to either the SPN or the DGC.

<https://doi.org/10.7554/eLife.103224.1.sa3>

Reviewer #2 (Public review):

Summary:

The authors have performed a rigorous study to assess the role of ESR1+ neurons in the PMC to control the coordination of bladder and sphincter muscles during urination. This is an important extension of previous work defining the role of these brainstem neurons, and convincingly adds to the understanding of their role as master regulators of urination. This is a thorough, well-done study that clarifies how the Pontine micturition center coordinates different muscle groups for efficient urination, but there are some questions and considerations that remain.

Strengths:

These data are thorough and convincing in showing that ESR1+ PMC neurons exert coordinated control over both the bladder and sphincter activity, which is essential for efficient urination. The anatomical distinctions in pelvic versus pudendal control are clear, and it's an advance to understand how this coordination occurs. This work offers a clearer picture of how micturition is driven.

Weaknesses:

The dynamics of how this population of ESR1+ neurons is engaged in natural urination events remains unclear. Not all ESR1+ neurons are always engaged, and it is not measured whether this is simply variation in population activity, or if more neurons are engaged during more intense starting bladder pressures, for instance. In particular, the response dynamics of single and doubly-projecting neurons are not defined. Additionally, the model for how these neurons coordinate with CRH+ neuron activity in the PMC is not addressed, although these cell types seem to be engaged at the same time. Lastly, it would be interesting to know how sensory input can likely modulate the activity of these neurons, but this is perhaps a future direction.

<https://doi.org/10.7554/eLife.103224.1.sa2>

Reviewer #3 (Public review):

Summary:

The paper by Li et al explored the role of Estrogen receptor 1 (Esr1) expressing neurons in the pontine micturition center (PMC), a brainstem region also known as Barrington's nucleus (Hou et al 2016, Keller et al 2018). First, the author conducted bulk Ca²⁺ imaging/unit recording from PMCESR1 to investigate the correlations of PMCESR1 neural activity to voiding behavior in conscious mice and bladder pressure/external urethral muscle activity in urethane anesthetized mice. Next, the authors conducted optogenetics inactivation/activation of PMCESR1 to confirm the contribution to the voiding behavior also conducted peripheral nerve transection together with optogenetics activation to confirm the independent control of bladder pressure and urethral sphincter muscle.

Weaknesses:

(1) The study demonstrates that pelvic nerve transection reduces urinary volume triggered by PMCESR1+ cell photoactivation in freely moving mice. Could the role of pudendal nerve transection also be examined in awake mice to provide a more comprehensive understanding of neural involvement?

- (2) While the paper primarily focuses on PMCESR1+ cells in bladder-sphincter coordination, the analysis of PMCESR1+-DGC/SPN neural circuits - given their distinct anatomical projections in the sacral spinal cord - feels underexplored. How do these circuits influence bladder and sphincter function when activated or inhibited? Also, do you have any tracing data to confirm whether bladder-sphincter innervation comes from distinct spinal nuclei?
- (3) Although the paper successfully identifies the physiological role of PMCESR1+ cells in bladder-sphincter coordination, the study falls short in examining the electrophysiological properties of PMCESR1+-DGC/SPN cells. A deeper investigation here would strengthen the findings.
- (4) The parameters for photoactivation (blue light pulses delivered at 25 Hz for 15 ms, every 30 s) and photoinhibition (pulses at 50 Hz for 20 ms) vary. What drove the selection of these specific parameters? Moreover, for photoactivation experiments, the change in pressure ($\Delta P = P5 \text{ sec} - P0 \text{ sec}$) is calculated differently from photoinhibition ($\Delta \text{pressure} = P_{\text{peak}} - P_{\text{min}}$). Can you clarify the reasoning behind these differing approaches?
- (5) The discussion could further emphasize how PMCESR1+ cells coordinate bladder contraction and sphincter relaxation to control urination, highlighting their central role in the initiation and suspension of this process.
- (6) In Figure 8, The authors analyze the temporal sequence of bladder pressure and EUS bursting during natural voiding and PMC activation-induced voiding. It would be acceptable to consider the existence of a lower spinal reflex circuit, however, the interpretation of the data contains speculation. Bladder pressure measurement is hard to say reflecting efferent pelvic nerve activity in real time. (As a biological system, bladder contraction is mediated by smooth muscle, and does not reflect real-time efferent pelvic nerve activity. As an experimental set-up, bladder pressure measurement has some delays to reflect bladder pressure because of tubing, but EUS bursting has no delay.) Especially for the inactivation experiment, these factors would contribute to the interpretation of data. This reviewer recommends a rewrite of the section considering these limitations. Most of the section is suitable for the results.

<https://doi.org/10.7554/eLife.103224.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

Urination requires precise coordination between the bladder and external urethral sphincter (EUS), while the neural substrates controlling this coordination remain poorly understood. In this study, Li et al. identify estrogen receptor 1-expressing neurons (ESR1+) in Barrington's nucleus as key regulators that faithfully initiate or suspend urination. Results from peripheral nerve lesions suggest that BarEsr1 neurons play independent roles in controlling bladder contraction and relaxation of the EUS. Finally, the authors performed region-specific retrograde tracing, claiming that distinct populations of BarEsr1 neurons target specific spinal nuclei involved in regulating the bladder and EUS, respectively.

Strength:

Overall, the work is of high quality. The authors integrate several cutting-edge technologies and sophisticated, thorough analyses, including opto-tagged single unit

recordings, combined optogenetics, and urodynamics, particularly those following distinct peripheral nerve lesions.

Weakness:

(1) My major concern is the novelty of this study. Keller et al. 2018 have shown that BarEsr1 neurons are active during urination and play an essential role in relaxing the external urethral sphincter (EUS). Minimally, substantial content that merely confirms previous findings (e.g. Figures 1A-E; Figures 3A-E) should be move to the supplementary datasets.

Indeed, we are aware of and have carefully studied the literature of Keller et al. Our manuscript here presents novel experiments beyond the scopes of that paper. Thanks to this comment, we will substantially revise our manuscript to enhance the visibility of novel data while keeping the agreeing data in the supplementary.

(2) I also have concerns regarding the results showing that the inactivation of BarEsr1 neurons led to the cessation of EUS muscle firing (Figures 2G and S5C). As shown in the cartoon illustration of Figure 8, spinal projections of BarEsr1 neurons contact interneurons (presumably inhibitory) that innervate motor neurons, which in turn excite the EUS. I would therefore expect that the inactivation of BarEsr1 should shift the EUS firing pattern from phasic (as relaxation) to tonic (removal of relaxation), rather than stopping their firing entirely. Could the authors comment on this and provide potential reasons or mechanisms for this finding?

We agree with this point. We meant that the EUS' phasic bursting pattern was rapidly stopped upon BarEsr1 photoinhibition, but not all the firing stopped instantaneously. According to the previous studies (Chang et al., 2007, de Groat, 2009, de Groat and Yoshimura, 2015, Kadekawa et al., 2016), the voiding physiology of rodents is probably different from that of humans, such that for rodents the urine is step-wise pumped out in the gap time between multiple consecutive EUS phasic bursting epochs, and for humans the urine is continuously pumped out once the EUS firing is almost fully inhibition during a period of time. Namely, for mice, the EUS display sustained tonic activity following phasic bursting, while, in contrast, for humans the EUS keeps tonic firing until the moment of voiding onset (complete inhibition, muscle relaxed). Despite the prominent differences in the basic physiological properties, our assumption is that the logic of circuits from the brainstem to the urethra in this pathway is evolutionally conserved for both species; thus the logic of brainstem coordination of voiding could also be the same for both species, which is the main interest of our study (of using an animal model to address concerns of human health). Thus, to interpret our data for a broader audience we made a simplified and inaccurate expression. We apologize for the inaccuracy and we will correct our previous inaccurate description in the revised manuscript.

(3) Current evidence is insufficient to support the claim that the majority of BarEsr1 neurons innervate the SPN but not DGC. The current spinal images are uninformative, as the fluorescence reflects the distribution of Esr1- or Crh-expressing neurons in the spinal cord, along with descending BarEsr1 or BarCrh axons. Given the close anatomical proximity of these two nuclei, a more thorough histological analysis is required to demonstrate that the spinal injections were accurately confined to either the SPN or the DGC.

We agree that current evidence is insufficient to support the current claim. To address this concern and strengthen our claim, we will repeat the retrograde viral tracing experiments, combined with CTB647 injections to label the injection site, to validate specific targeting of SPN or DGC populations. We will also add higher-magnification imaging to distinguish

BarESR1 axonal projections targeting SPN versus DGC. Results from these ongoing experiments will be incorporated into the revised manuscript.

Reviewer #2 (Public review):

Summary:

The authors have performed a rigorous study to assess the role of ESR1+ neurons in the PMC to control the coordination of bladder and sphincter muscles during urination. This is an important extension of previous work defining the role of these brainstem neurons, and convincingly adds to the understanding of their role as master regulators of urination. This is a thorough, well-done study that clarifies how the Pontine micturition center coordinates different muscle groups for efficient urination, but there are some questions and considerations that remain.

Strengths:

These data are thorough and convincing in showing that ESR1+PMC neurons exert coordinated control over both the bladder and sphincter activity, which is essential for efficient urination. The anatomical distinctions in pelvic versus pudendal control are clear, and it's an advance to understand how this coordination occurs. This work offers a clearer picture of how micturition is driven.

Weaknesses:

The dynamics of how this population of ESR1+ neurons is engaged in natural urination events remains unclear. Not all ESR1+neurons are always engaged, and it is not measured whether this is simply variation in population activity, or if more neurons are engaged during more intense starting bladder pressures, for instance. In particular, the response dynamics of single and doubly-projecting neurons are not defined. Additionally, the model for how these neurons coordinate with CRH+ neuron activity in the PMC is not addressed, although these cell types seem to be engaged at the same time. Lastly, it would be interesting to know how sensory input can likely modulate the activity of these neurons, but this is perhaps a future direction.

In response to the reviewer's comments, we will attempt perform the following revisions for this round:

(1) Engagement of ESR1+ neurons in natural urination events:

We agree that probably not all ESR1+ neurons are consistently engaged during urination. To address this, we will perform a detailed analysis of the opto-tagged single unit recordings data.

(2) Response dynamics of single- and doubly-projecting neurons:

(a) We will use retrograde labelling combined with Ca²⁺ photometry recordings to differentiate the response dynamics of SPN- and DGC-projecting neurons during urination.

(b) We will perform functional validations to assess the specific roles of single- and doubly-projecting neurons in coordinating bladder and EUS activity.

(3) Coordination with CRH+ neurons in the PMC:

We appreciate the suggestion to include CRH+ neurons in our model. We will expand our model to incorporate CRH+ neurons and their potential interactions with ESR1+ neurons.

(4) Sensory modulation of ESR1+ neurons:

The reviewer raises an excellent point regarding sensory input modulation of ESR1+ neuron

activity. Although this is beyond the scope of our current study, we recognize its importance and propose to include this as a future direction.

Reviewer #3 (Public review):

Summary:

The paper by Li et al explored the role of Estrogen receptor 1 (Esr1) expressing neurons in the pontine micturition center (PMC), a brainstem region also known as Barrington's nucleus (Hou et al 2016, Keller et al 2018). First, the author conducted bulk Ca²⁺ imaging/unit recording from PMCESR1 to investigate the correlations of PMCESR1 neural activity to voiding behavior in conscious mice and bladder pressure/external urethral muscle activity in urethane anesthetized mice. Next, the authors conducted optogenetics inactivation/activation of PMCESR1 to confirm the contribution to the voiding behavior also conducted peripheral nerve transection together with optogenetics activation to confirm the independent control of bladder pressure and urethral sphincter muscle.

Weaknesses:

(1) The study demonstrates that pelvic nerve transection reduces urinary volume triggered by PMCESR1+ cell photoactivation in freely moving mice. Could the role of pudendal nerve transection also be examined in awake mice to provide a more comprehensive understanding of neural involvement?

Thank you for the suggestion, the pudendal nerve transection in awake mice is indeed a challenging experiment that has been missed. We will try it for the revision.

(2) While the paper primarily focuses on PMCESR1+ cells in bladder-sphincter coordination, the analysis of PMCESR1+-DGC/SPN neural circuits - given their distinct anatomical projections in the sacral spinal cord - feels underexplored. How do these circuits influence bladder and sphincter function when activated or inhibited? Also, do you have any tracing data to confirm whether bladder-sphincter innervation comes from distinct spinal nuclei?

Thank you for this great comment. The projection-specific neuronal function analysis is, as also suggested by Reviewer 2 in a similar comment (#8), missing in our first submission. These are so challenging experiments that we have missed in the first round of tests, but we decide to pursue this goal again. Namely, we will perform photometry recordings of PMC neurons projecting to the DGC/SPN during measuring bladder pressure and urethral sphincter EMG activity. Additionally, while our study does not include direct tracing data to confirm distinct spinal nuclei for bladder and sphincter innervation, this has been well-documented in classic literature (Yao et al., 2018, Karnup and De Groat, 2020, Karnup, 2021). Specifically, anatomical studies have shown that SPN primarily innervates the bladder, while the DGC is associated with the innervation of the urethral sphincter. We will cite these references to provide context and support for our interpretations.

(3) Although the paper successfully identifies the physiological role of PMCESR1+ cells in bladder-sphincter coordination, the study falls short in examining the electrophysiological properties of PMCESR1+-DGC/SPN cells. A deeper investigation here would strengthen the findings.

While our study primarily focuses on the functional role of PMCESR1+ neurons in bladder-sphincter coordination, we acknowledge that understanding their intrinsic electrophysiological characteristics could further strengthen our findings. However, this aspect falls beyond the scope of the current study. Nevertheless, we recognize the significance

of this direction and are excited to pursue it in future research. We appreciate the reviewer's suggestion, as it highlights an important avenue for expanding upon our current findings.

(4) The parameters for photoactivation (blue light pulses delivered at 25 Hz for 15 ms, every 30 s) and photoinhibition (pulses at 50 Hz for 20 ms) vary. What drove the selection of these specific parameters? Moreover, for photoactivation experiments, the change in pressure ($\Delta P = P5 \text{ sec} - P0 \text{ sec}$) is calculated differently from photoinhibition ($\Delta \text{pressure} = P_{\text{peak}} - P_{\text{min}}$). Can you clarify the reasoning behind these differing approaches?

We sincerely thank the reviewer for raising these important points and for the opportunity to clarify our experimental design and data analysis methods.

Photoactivation versus photoinhibition parameters: The differences in photoactivation (25 Hz, 15 ms pulses) and photoinhibition (50 Hz, 20 ms pulses) protocols are based on the distinct physiological and technical requirements for activating versus inhibiting PMCESR1+ neurons. For photoactivation, 25 Hz stimulation aligns with the natural firing patterns of central neurons, allowing for intermittent activation without exceeding the neuronal refractory period. The shorter pulse duration (15 ms) minimizes phototoxicity and avoids overstimulation, as performed in previous studies (Keller et al., 2018). In contrast, photoinhibition requires sustained suppression of neuronal activity, achieved through higher frequencies (50 Hz) and longer pulses (20 ms) to ensure continuous coverage of neuronal activity.

Calculation of pressure changes (ΔP) for photoactivation and photoinhibition: The differing methods for calculating pressure changes reflect the distinct physiological effects we aimed to capture. In photoactivation experiments ($\Delta P = P5 \text{ sec} - P0 \text{ sec}$), the pressures before ($P0 \text{ sec}$) and 5 seconds after ($P5 \text{ sec}$) light delivery were compared to capture the immediate effect of light activation on bladder pressure, focusing on the onset and early dynamics of activation. In contrast, photoinhibition experiments assessed the immediate impact of light-induced suppression on bladder pressure during an ongoing voiding event. Here, $\Delta \text{pressure}$ was calculated as $P_{\text{peak}} - P_{\text{min}}$ to measure the rapid drop in pressure directly attributable to neuronal inhibition.

We will expand these details in the methods section of the revised manuscript to provide greater transparency.

(5) The discussion could further emphasize how PMCESR1+ cells coordinate bladder contraction and sphincter relaxation to control urination, highlighting their central role in the initiation and suspension of this process.

We fully agree with this point. Additionally, in response to your and other reviewers' suggestions, we are preparing a new round of experiments with projection-specific recording, and thus our discussion and conclusion will also be updated according to the newly obtained data.

(6) In Figure 8, The authors analyze the temporal sequence of bladder pressure and EUS bursting during natural voiding and PMC activation-induced voiding. It would be acceptable to consider the existence of a lower spinal reflex circuit, however, the interpretation of the data contains speculation. Bladder pressure measurement is hard to say reflecting efferent pelvic nerve activity in real time. (As a biological system, bladder contraction is mediated by smooth muscle, and does not reflect real-time efferent pelvic nerve activity. As an experimental set-up, bladder pressure measurement has some delays to reflect bladder pressure because of tubing, but EUS bursting has no delay.) Especially for the inactivation experiment, these factors would contribute to the

interpretation of data. This reviewer recommends a rewrite of the section considering these limitations. Most of the section is suitable for the results.

Thank you for mentioning the possibility of bladder pressure measurement delay. We would prefer to perform a physical control test to quantify how much delay this measurement is under our experimental conditions. We will use a small balloon to mimic the bladder and use two identical pressure sensors, one with a very short tube inserted into the balloon and one with an extended tube same as in our animal experiments. We will then mimic both contraction initiation and halting, and quantify the delay between the two sensors.

References

- Chang HY, Cheng CL, Chen JJJ, de Groat WC. 2007. Serotonergic drugs and spinal cord transections indicate that different spinal circuits are involved in external urethral sphincter activity in rats. *American Journal of Physiology-Renal Physiology* 292: F1044-F1053. DOI: 10.1152/ajprenal.00175.2006
- de Groat WC. 2009. Integrative control of the lower urinary tract: preclinical perspective. *British Journal of Pharmacology* 147. DOI: 10.1038/sj.bjp.0706604
- de Groat WC, Yoshimura N. 2015. Anatomy and physiology of the lower urinary tract. *Handb Clin Neurol* 130: 61-108. DOI: 10.1016/B978-0-444-63247-0.00005-5
- Kadekawa K, Yoshimura N, Majima T, Wada N, Shimizu T, Birder LA, Kanai AJ, de Groat WC, Sugaya K, Yoshiyama M. 2016. Characterization of bladder and external urethral activity in mice with or without spinal cord injury—a comparison study with rats. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* 310: R752-R758. DOI: 10.1152/ajpregu.00450.2015
- Karnup S. 2021. Spinal interneurons of the lower urinary tract circuits. *Autonomic Neuroscience* 235. DOI: 10.1016/j.autneu.2021.102861
- Karnup SV, De Groat WC. 2020. Mapping of spinal interneurons involved in regulation of the lower urinary tract in juvenile male rats. *IBRO Rep* 9: 115-131. DOI: 10.1016/j.ibror.2020.07.002
- Keller JA, Chen J, Simpson S, Wang EH-J, Lilascharoen V, George O, Lim BK, Stowers L. 2018. Voluntary urination control by brainstem neurons that relax the urethral sphincter. *Nature Neuroscience* 21: 1229-1238. DOI: 10.1038/s41593-018-0204-3
- Yao J, Zhang Q, Liao X, Li Q, Liang S, Li X, Zhang Y, Li X, Wang H, Qin H, Wang M, Li J, Zhang J, He W, Zhang W, Li T, Xu F, Gong H, Jia H, Xu X, Yan J, Chen X. 2018. A corticopontine circuit for initiation of urination. *Nature Neuroscience* 21: 1541-1550. DOI: 10.1038/s41593-018-0256-4

<https://doi.org/10.7554/eLife.103224.1.sa0>